

MQCT 2026 User Guide

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²The most recent version is always available at GitHub: <https://github.com/MarquetteQuantum/MQCT>

I. Molecular System Types and Rotation-Vibration Wavefunctions Included in MQCT

In MQCT program there are ten system types, summarized in Table 1, from the simplest rigid-diatom + atom, to the most general case of two asymmetric-top rotor molecules. For each system, MQCT calculations can be initiated by indicating the rotational and vibrational constants listed in the Table. Those are used by the code to set up and diagonalize Hamiltonian matrix for rotational motion (using the basis set of symmetric-top functions) in order to determine the rotational states of the system (energies and wave-functions).

Table 1: Ten types of systems handled by MQCT, with required and optional input data

SYS_TYPE	Collision Partners	Required Constants	Channel Labels	Optional Input
1	rigid diatom + atom	B_e, D_e	J	
2	vibrating diatom + atom	B_e, D_e, ω_e, x_e	j, v	vibrational functions, non-equidistant grid
3	symmetric top + atom	A, C	j, k, ε	
4	asymmetric top + atom	A, B, C	j, k_a, k_c	expansion over sym. top basis
5	rigid diatom + rigid diatom	$B_{e1}, D_{e1},$ B_{e2}, D_{e2}	j_1, j_2	
6	vibrating diatom + vibrating diatom	$B_{e1}, D_{e1}, \omega_{e1}, x_{e1},$ $B_{e2}, D_{e2}, \omega_{e2}, x_{e2}$	j_1, v_1, j_2, v_2	vibrational functions, non-equidistant grid
7	symmetric top + rigid diatom	$A, C,$ B_e, D_e	$j_1, k_1, \varepsilon_1, j_2$	
8	asymmetric top + rigid diatom	$A, B, C,$ B_e, D_e	j_1, k_{a1}, k_{c1}, j_2	expansion over sym. top basis
9	asymmetric top + symmetric top	$A_1, B_1, C_1,$ A_2, C_2	$j_1, k_{a1}, k_{c1}, j_2, k_2, \varepsilon_2$	expansion over sym. top basis
0	asymmetric top + asymmetric top	$A_1, B_1, C_1,$ A_2, B_2, C_2	$j_1, k_{a1}, k_{c1}, j_2, k_{a2}, k_{c2}$	expansion over sym. top basis

For a general asymmetric top molecule, the values of A , B and C should correspond to the x , y , and z -axes (not necessarily in decreasing order) and must be consistent with reference orientation of the potential energy surface for the system (*i.e.*, all Euler angles set to zero). If the molecule has a symmetry axis, user can take advantage of this, as usual, by choosing the reference orientation such that the symmetry axis is aligned with z , to enable independent calculations for para- and ortho-states of the molecule (and thus to reduce the rotational basis set size, accordingly). When symmetric tops are involved in collisions, the case of an oblate top is handled in a standard way, with rotational constants indicated such that $A > C$, while for a prolate top the input should be in the form $A < C$, opposite to the standard notation. In both cases it is assumed that $B = A$. Although the case of spherical top is not explicitly included, it can also be handled, by entering equal values for A and C .

The option of invoking the externally computed user-supplied rotational-vibrational states (*e.g.*, vibrational wave functions defined on a non-equidistant grid of points, or vibrationally distorted rotational states, such as those of Kyro model Hamiltonian, expanded over basis set of symmetric-top functions) is indicated in Table 1, where available, and is described in more detail in Sec. VI below (see Table 7).

Explicit form of rotational wavefunction constructed by MQCT is given in Table 2 for all system types. The ranges of quantum numbers are:

$$\begin{aligned} |j_1 - j_2| \leq j \leq j_1 + j_2, & \quad 0 \leq k_1 \leq j_1 \text{ (positive only),} & \quad \varepsilon_1 = \pm \text{ (parity),} \\ -j \leq m \leq +j, & \quad 0 \leq k_2 \leq j_2 \text{ (positive only),} & \quad \varepsilon_2 = \pm \text{ (parity).} \end{aligned}$$

For symmetric top molecules ($\text{SYS_TYPE}=3, 7, 9$) the input values of *inversion parity* are $\varepsilon = 0$ for positive parity (+) and $\varepsilon = 1$ for negative parity (−), since in the code the parity is implemented as a phase factor $(-1)^\varepsilon$. For $k = 0$ only positive parity is allowed ($\varepsilon = 0$). Also note that for asymmetric top molecules ($\text{SYS_TYPE}=4, 8, 9, 0$) there are $2j + 1$ states of the type $j_{k_a k_c}$ with specific sets of coefficients b_k , for each molecule (expansion over the basis set of symmetric-top eigenstates). These are automatically computed by the code or, alternatively, users can supply their own sets of coefficients b_k pre-computed externally (*e.g.*, for Kyro-type Hamiltonian) for one ($\text{SYS_TYPE}=4, 8, 9$) or both ($\text{SYS_TYPE}=0$) asymmetric top collision partners (see Sec. VI, Table 7 and example input files).

Vibrational wavefunctions of diatomics ($\text{SYS_TYPE}=2, 6$) are defined on a grid of points, equidistant by default but, a user-supplied non-equidistant grid can also be used (see Sec. VI, Table 7). In both cases the integrals of vibrational wavefunctions include the weights of grid points. MQCT code will first try to search for the file `USER_DEFINED_BASIS.DAT` which contains energies, vibrational wavefunctions, along with the weights of grid points, for one ($\text{SYS_TYPE}=2$) or both ($\text{SYS_TYPE}=6$) collision partners, see Sec. VI. If not provided, MQCT would use Morse parameters from the input file to construct the vibrational wavefunctions of non-rotating Morse oscillator. If those are not provided, it would compute the values of Morse parameters from spectroscopic coefficients indicated in the input file. Vibrational energies are computed using a standard Dunham’s formula. Vibrational wave functions of Morse oscillator ($j = 0$, neglecting ro-vibrational distortion) are computed using recurrent relations and are used to compute matrix elements for the corresponding state-to-state transitions.

Table 2: Wavefunctions of rotational-vibrational eigenstates in MQCT calculations of different system types

System type	Representation of Wavefunctions
1	$ jm\rangle = Y_m^j(\beta, \alpha)$
2	$ jmv\rangle = \psi_v(r) \times Y_m^j(\beta, \alpha)$
3	$ jmk\varepsilon\rangle = \sqrt{\frac{2j+1}{8\pi^2}} \sqrt{\frac{1}{2(1+\delta_{k,0})}} [D_{m,+k}^{j*}(\alpha, \beta, \gamma) + \varepsilon D_{m,-k}^{j*}(\alpha, \beta, \gamma)]$
4	$ jm\rangle = \sqrt{\frac{2j+1}{8\pi^2}} \left(\sum_{k=-j}^{+j} b_k D_{m,k}^{j*}(\alpha, \beta, \gamma) \right)$
5	$ jmj_1j_2\rangle = \sum_{m_1=-j_1}^{+j_1} C_{j_1,m_1,j_2,m-m_1}^{j,m} Y_{m_1}^{j_1}(\beta_1, \alpha_1) \times Y_{m-m_1}^{j_2}(\beta_2, \alpha_2)$
6	$ jmj_1j_2v_1v_2\rangle = \psi_{v_1}(r_1) \times \psi_{v_2}(r_2) \times \sum_{m_1=-j_1}^{+j_1} C_{j_1,m_1,j_2,m-m_1}^{j,m} Y_{m_1}^{j_1}(\beta_1, \alpha_1) \times Y_{m-m_1}^{j_2}(\beta_2, \alpha_2)$
7	$ jmj_1k_1\varepsilon_1j_2\rangle = \sqrt{\frac{2j_1+1}{8\pi^2}} \sqrt{\frac{1}{2(1+\delta_{k_1,0})}} \sum_{m_1=-j_1}^{+j_1} C_{j_1,m_1,j_2,m-m_1}^{j,m} [D_{m_1,+k_1}^{j_1*}(\alpha_1, \beta_1, \gamma_1) + \varepsilon_1 D_{m_1,-k_1}^{j_1*}(\alpha_1, \beta_1, \gamma_1)] \times Y_{m-m_1}^{j_2}(\beta_2, \alpha_2)$
8	$ jmj_1j_2\rangle = \sqrt{\frac{2j_1+1}{8\pi^2}} \sum_{m_1=-j_1}^{+j_1} C_{j_1,m_1,j_2,m-m_1}^{j,m} \left(\sum_{k_1=-j_1}^{+j_1} b_{k_1} D_{m_1,k_1}^{j_1*}(\alpha_1, \beta_1, \gamma_1) \right) \times Y_{m-m_1}^{j_2}(\beta_2, \alpha_2)$
9	$ jmj_1j_2k_2\varepsilon_2\rangle = \sqrt{\frac{2j_1+1}{8\pi^2}} \sqrt{\frac{2j_2+1}{8\pi^2}} \sqrt{\frac{1}{2(1+\delta_{k_2,0})}} \sum_{m_1=-j_1}^{+j_1} C_{j_1,m_1,j_2,m-m_1}^{j,m} \left(\sum_{k_1=-j_1}^{+j_1} b_{k_1} D_{m_1,k_1}^{j_1*}(\alpha_1, \beta_1, \gamma_1) \right) \times [D_{m-m_1,+k_2}^{j_2*}(\alpha_2, \beta_2, \gamma_2) + \varepsilon_2 D_{m-m_1,-k_2}^{j_2*}(\alpha_2, \beta_2, \gamma_2)]$
0	$ jmj_1j_2\rangle = \sqrt{\frac{2j_1+1}{8\pi^2}} \sqrt{\frac{2j_2+1}{8\pi^2}} \sum_{m_1=-j_1}^{+j_1} C_{j_1,m_1,j_2,m-m_1}^{j,m} \left(\sum_{k_1=-j_1}^{+j_1} b_{k_1} D_{m_1,k_1}^{j_1*}(\alpha_1, \beta_1, \gamma_1) \right) \times \left(\sum_{k_2=-j_2}^{+j_2} b_{k_2} D_{m-m_1,k_2}^{j_2*}(\alpha_2, \beta_2, \gamma_2) \right)$

In the code we use Euler angles that correspond to active rotations of each molecule, according to the intrinsic $z - y' - z''$ convention. The expression for Wigner D-function is $D_{mk}^j(\alpha, \beta, \gamma) = e^{-im\alpha} d_{mk}^j(\beta) e^{-ik\gamma}$, where j is angular momentum, m is its projection onto z -axis (that connects centers of mass of two collision partners), and k is projection of j onto the molecule-fixed axis z' (such as symmetry axis, or one of the principal axis of inertia). Eigenfunction of symmetric top is D_{mk}^{j*} , with eigenvalues of the corresponding operators given by $\hat{J}^2 D_{mk}^{j*} = j(j+1) D_{mk}^{j*}$, $\hat{J}_{z'} D_{mk}^{j*} = k D_{mk}^{j*}$, and $\hat{J}_z D_{mk}^{j*} = m D_{mk}^{j*}$. Eigenfunction of a linear top is $Y_m^j = \sqrt{(2j+1)/(4\pi)} D_{m0}^{j*}$.

II. Structure of the Input File, Required and Optional Keywords, Default Values

Initial conditions for classical and quantum degrees of freedom in the system are defined separately in two blocks of the input file, called \$SYSTEM and \$BASIS. Potential energy surface describes interaction between quantum and classical parts of the system and is defined in the third block of the input file, called \$POTENTIAL. This block should be the last in the input file, while the order of the first two blocks is interchangeable since they are independent. Example below sets up the input for calculations of $\text{H}_2\text{O} + \text{He}$ rotational excitation (a complete list of required and optional keywords is given in Tables 3, 4 and 5 below):

```
$BASIS
  SYS_TYPE=4, A=27.877, B=9.285, C=14.512,
  NMB_CHNLS=6,
  CHNLS_LIST=0,0,0, 1,0,1, 1,1,0, 1,1,1, 2,0,2, 2,2,1,
  INIT_CHNL=0,0,0
$END

$SYSTEM
  LABEL="H2O+He", MASS_RED=3.2748, RMIN=4.5, RMAX=20.0,
  NMB_ENERGS=1, U_ENERGY=200., JTOTL=0, JTOTU=100,
  TIME_STEP=15.0, TIME_LIM=3.5E+6
$END

$POTENTIAL
  E_UNITS=A.U., R_UNITS=A.U.,
  GRD_R=50, GRD_ANG1=40, GRD_ANG2=20, GRD_ANG3=40
$END
```

Each block starts with its name and is finalized by the \$END. All entries inside the block are separated by coma. You can have as many spaces, lines or tabs as you want between the keywords and their values, all is taken care of by the parser. First keyword of the block \$BASIS indicates the type of the system, followed by three values of the rotational constants of the molecule (in the units of wavenumber). Here the rotational states of the asymmetric top rotor are labelled using the standard notation: j, k_a, k_c (where k_a and k_c are projections of j onto the principal axis with smallest and largest moments of inertia, respectively). Another useful optional keyword allows forming a basis set out of all states below given energy, for example: EMAX=135.4 commands to include in the basis set all states at energies below 135.4 cm⁻¹ (which, again, would be the same six channels). Rotational quenching calculations can be initiated by choosing an excited state as initial, for example: INIT_CHNL=2, 0, 2.

The block \$SYSTEM, besides the text LABEL, contains reduced mass of the collision partners (MASS_RED, in atomic mass units), the minimum and maximum values of distance between collision partners to initialize and terminate trajectories (RMIN and RMAX in the units of Bohr), the number of collision energies to compute (NMB_ENERGS) and their effective values (U_ENERGY in wavenumbers), the range of total angular momentum quantum number J (from JTOTL to JTOTU), propagation time-step TIME_STEP and the time limit TIME_LIM to terminate trajectories (both in atomic units). One useful option is to set the maximum value of impact parameter, in the units of Bohr, for example: B_IMPCT=9.50. If this is specified, then the upper and lower limits of J are not required (JTOTL and JTOTU are ignored, even if indicated). This option is convenient for calculations in a broad energy range, since the maximum impact parameter is less sensitive to collision energy. At each collision energy, the upper limit of J is determined individually, based on the indicated value of maximum impact parameter. By default, 4th-order Runge-Kutta method is employed to propagate MQCT trajectories (together with quantum equations for state populations) using a constant step-size TIME_STEP. Trajectories in MQCT are integrated through the interaction region and are terminated when the molecule-molecule distance exceeds RMAX. If optional time-limit is indicated (by TIME_LIM, as in the example above) the trajectory is terminated as soon as either condition is fulfilled. This is recommended at low collision energies, when orbiting trajectories are possible (analogous to quantum scattering resonances, see below).

In the block \$POTENTIAL, the units of energy and distance for the potential energy surface subroutine (supplied by the user, see below) are indicated first, followed by the number of quadrature points for integration of matrix elements for state-to-state transitions. For the units of distance, (keyword R_UNITS), Bohr and Angstrom are available, as defined by A.U. and ANGS, respectively. For energy units (keyword E_UNITS) possible values are A.U., CM-1, KLVN and KCAL that correspond to Hartree, wavenumber, Kelvin and kilocalorie per mole, respectively. Note that angles are always assumed to be in radians. By default, state-to-state transition matrix elements are computed by direct numerical integration over all internal degrees of freedom. The number of quadrature points should be indicated for each angular coordinate, as in the example above (three Euler angles for an asymmetric top rotor, such as H₂O). Integration over angles is carried out at each point of the molecule-molecule distance grid (GRD_R points in the range between RMIN and RMAX). In the collision dynamics calculations, each matrix element is splined over the distance between the grid points, using 3rd-order one-dimensional spline. Optionally, computed matrix elements can be saved to a file, using keyword SAVE_MTRX=YES. By default, this

would be in the binary form (unformatted). Formatted matrix output can be requested by the keyword UNFORMAT=NO, *e.g.*, for the inspection by the user.

The three tables below give a comprehensive list of all required and optional input parameters for three parts of the input file, the blocks \$BASIS, \$SYSTEM, and \$POTENTIAL. Default values are indicated where applicable. A keyword must be specified only if a value different from the default is desired. Datatype “real” corresponds to double precision. The values “YES” and “NO” correspond to logical datatype.

Table 3: Description of keywords for the block \$BASIS

Keyword	Type, Range, Units	Description	Relevant SYS_TYPE
SYS_TYPE	integer, 0 to 9	1 -- rigid diatom + atom 2 -- vibrating diatom + atom 3 -- symmetric top + atom 4 -- asymmetric top + atom 5 -- diatom + diatom (both rigid) 6 -- vibrating diatom + vibrating diatom 7 -- symmetric top + diatom (rigid) 8 -- asymmetric top + diatom (rigid) 9 -- asymmetric top + symmetric top 0 -- asymmetric top + asymmetric top	
NMB_CHNLS	Integer	Number of channels	all
CHNLS_LIST	Integers	Quantum numbers to specify channels	all
INIT_CHNL	Integers	Quantum numbers to specify the initial channel	all
EXCLUDE_STATES	default is “NO”	User can exclude specified states from the basis set, <i>e.g.</i> , the weakly coupled states or the states with certain <i>j</i> and <i>m</i> .	all
BE, DE	real, positive, cm ⁻¹	Rotational constants B_e, D_e (exp. format for D_e)	1,2,7,8
A, B, C	real, positive, cm ⁻¹	Rotational constants A, B, C (for <i>x, y</i> and <i>z</i> , as in PES)	3,4,7,8
BE1, DE1	real, positive, cm ⁻¹	Rotational constants B_e, D_e for molecule #1	5,6
BE2, DE2	real, positive, cm ⁻¹	Rotational constants B_e, D_e for molecule #2	5,6,7,8
A1, B1, C1	real, positive, cm ⁻¹	Rotational constants A, B, C for molecule #1 (for <i>x, y</i> and <i>z</i>)	9,0
A2, B2, C2	real, positive, cm ⁻¹	Rotational constants A, B, C for molecule #2 (for <i>x, y</i> and <i>z</i>)	9,0
WE, XE	real, positive, cm ⁻¹	Vibrational constants ω_e, x_e	2

WE1, XE1	real, positive, cm^{-1}	Vibrational constants ω_e, x_e for molecule #1	6
WE2, XE2	real, positive, cm^{-1}	Vibrational constants ω_e, x_e for molecule #2	6
JMIN, JMAX	integer	The range of rotational number j included in the basis set; optional, used to avoid listing all levels individually; can use with EMAX.	1,2,3,4
VMIN, VMAX	integer	The range of vibrational number v included in basis set; optional, used to avoid listing all levels individually	2
JMIN1, JMAX1, JMIN2, JMAX2	integer	The range of rotational numbers included in basis sets for molecules #1, 2; can use with EMAX or with EMAX1, EMAX2	5,6,7,8,9,0
VMIN1, VMAX1, VMIN2, VMAX2	integer	The range of vibrational numbers included in basis sets for molecules #1, 2; option, used to avoid listing the levels	6
NCHL1, NCHL2	integer	Number of lower energy channels included for molecules #1, 2; optional, used to avoid listing the levels individually	5,6,7,8,9,0
EMAX	real, cm^{-1}	Channel energy cutoff. Only the states below are included in calculations; option, to avoid listing all levels	all
EMAX1, EMAX2	real, cm^{-1}	Channel energy cutoffs for molecules #1 and #2; only the states below this energy are included in calculations	5,6,7,8,9,0
SYMMETRY	default is "NO"	If "YES", only the states coupled to the initial state are retained in the basis (ortho vs para states).	3,4,7,8,9,0
ATOMIC_MASSES	real, positive, amu	Masses of atoms in the diatomic; to determine COM	2,6
MORSE_DEPTH	real, positive, cm^{-1}	Depth parameter of Morse oscillator (dissociation energy)	2,6
MORSE_WIDTH	real, positive, Bohr	Width parameter of Morse oscillator	2,6
MORSE_POSITN	real, positive, Bohr	Equilibrium Distance of Morse oscillator	2,6
RMIN_VIBGRID	real, positive, Bohr	Minimum diatomic bond length	2,6
RMAX_VIBGRID	real, positive, Bohr	Maximum diatomic bond length	2,6
WGHT_POSPAR	real, positive, ≤ 1.0	Weight of positive total parity wave functions in the case of identical particles collision. Default is 1.0	5,6,0
CHNL_ENERGS	real, cm^{-1}	User can list energies of states (e.g., computed externally)	all
LEVELS_FILE	default is "NO"	If "YES", user provides energies, wave functions and assignments of externally-computed states, in a file	all
CS_APPROX	default is "NO"	If "YES", the coupled-states approximation will be used	all
IDENTICAL	default is "NO"	If "YES", collision partners are treated as indistinguishable	5,6,0
PRINT_STATES	default is "NO"	If "YES", prints out states and structure of coupling matrix	all
SAVE_TRAJECT	default is "NO"	If "YES", trajectory info is saved to files for AT-MQCT	all
AT_APPROX	default is "NO"	If "YES", AT-MQCT is done using saved trajectory info	all

Table 4: Description of keywords for the block \$SYSTEM

Keyword	Type, Range, Units	Description
LABEL	text	Name of your job
MASS_RED	real, positive, amu	Reduced mass of two scattering partners
RMIN, RMAX	real, positive, Bohr	Minimum and maximum values of distance between partners
B_IMPCT	real, positive, Bohr	Maximum value of collision impact parameter
JTOTL, JTOTU	integer	Lower and upper limits of total angular momentum J
NMB_ENERGS	integer	Number of collision energy values U to propagate
U_ENERGY	real, positive, cm^{-1}	List of collision energies U to propagate
UMIN, UMAX	real, positive, cm^{-1}	Minimum and maximum collision energies U
DU	real, positive, cm^{-1}	Step size for setting collision energies U
TIME_STEP	real, positive, au	Propagation time step for RK4, or maximum time step allowed for ODEINT
MIN_TMSTP	real, positive, au	Minimum time step allowed in ODEINT
SINGLE_STEP	default is “NO”	Propagates each trajectory in one step using ODEINT
TIME_LIM	real, positive, au	Time limit for propagation
EPS_ODEINT	real, positive, <1.d0	Relative error for step-size control in ODEINT (exp. format)
EPS_MONCAR	real, positive, %	Desirable error in Monte Carlo sampling of initial conditions
PROPAGATOR	text	RK4 is default, ODEINT is optional
MPI_PERTRAJ	integer, default is 1	Number of CPUs (MPI tasks) to use for calculations of each trajectory
NMB_LOOPS	integer, default is 1	Number of full loops (360 deg.) to propagate for orbiting trajectories
NMB_OSCIL	integer, default is 1	Number of outer turning points to propagate for oscillating trajectories
NO_RESONANCE	default is “NO”	If “YES”, orbiting trajectories are removed from analysis
DIFF_CROSS	default is “NO”	If “YES”, differential cross section is computed (elastic only)
ANG_RES	integer, default is 1000	Number of points for angular resolution of the differential cross section
MONTE_CARLO	default is “NO”	If “YES”, initial conditions are sampled randomly
NMB_TRAJ	integer, default is 100	Number of trajectories to sample using Monte Carlo (total number of trajectories)
CHECK_POINT	integer	Wall clock time (minutes after the start) to start writing a checkpoint file
RESTART	default is “NO”	If “YES”, program will start from a check point file
DL	integer, default is 1	Step size for orbital angular momentum ℓ ($DL > 1$ is used for speedup)
DL_MR	integer, default is 1	DL for the mid-range part of the interaction potential (use $DL_MR=1$ for accuracy)
DL_LR	integer, default is 1	DL for the long-range part of the interaction potential (normally $DL_LR > DL$)
B_MR	real, positive	Impact parameter (in Bohr) where the sampling switches from DL to DL_MR
B_LR	real, positive	Impact parameter (in Bohr) where the sampling switches from DL or DL_MR to DL_LR

PROB_SPLINE	default is “NO”	If “YES”, transition probabilities are interpolated to obtain the values skipped by DL
PRN_TRJCT	integer	Indicates the value of ℓ for which all the trajectory data will be printed
PRN_JM	two integers, coma separated	Indicates the values of j and m for PRN_TRJCT and PRN_SPLINE, defaults are $j = \max\{j_1, j_2\}$ and $m = 0$
PRN_P	integer, default is 0 for (+)	Indicates the desired parity (+ or −) for PRN_TRJCT in the case of identical collision partners
PRN_AMPLTD	default is “NO”	If “YES”, prints Re and Im parts of all probability amplitudes at the final moment of time
PRN_MTRX	default is “NO”	If “YES”, prints matrix elements indicated by TRANS_LST along trajectory given by PRN_TRJCT
TRANS_LST	integer	Specifies the list of transitions to be printed to the output file by PRN_MTRX.
REF_XZ	default is “YES”	Should be set to “NO” if the symmetry/reference plane of the PES is other than XZ-plane
ENERGY_ERROR	default is “YES”	“NO” disables calculations of energy conservation error (for speedup, CS or AT-MQCT)
PRN_PES	default is “NO”	If YES, a slice of the PES defined by FIX_VAR is saved to a file V_Slice_2D.out
FIX_VAR	9 reals, coma separated	PES variables: $R, r_1, r_2, \alpha_1, \beta_1, \gamma_1, \alpha_2, \beta_2, \gamma_2$. Use -1.0 for those two you wish to vary, fix the rest
AT_ADAPTOL	real, positive	If indicated, RK4 with adaptive step size is used for AT-MQCT, with this tolerance
POLAR_ANGLE	default is “NO”	If “Yes”, right hand side is computed and the equation of motion for Θ is propagated

Table 5: Description of keywords for the block \$POTENTIAL

Keyword	Type, Range, Units	Description	Relevant SYS_TYPE
READ_MTRX	default is “NO”	If “YES”, read the potential coupling matrix from file	all
SAVE_MTRX	default is “NO”	If “YES”, write the potential coupling matrix to file	all
UNFORMAT	default is “YES”	Saves matrix in binary form; set “NO” to save it as text	all
PROG_RUN	default is “YES”	Propagates trajectories; set “NO” to compute matrix only	all
E_UNITS	Text	Energy units of PES: “A.U.”, “CM-1”, “KCAL” or “KLVN”	all
R_UNITS	Text	Distance units for supplied PES: “A.U.” or “ANGS”	all
GRD_R	Integer	Number of points for R -grid	all
GRD_VIB	integer / two integers	Number of points for vibrational grid / grids	2 / 6
GRD_ANG1	integer / two integers	Number of points for α -grid / grids	1,2,3,4 / 5,6,7,8,9,0
GRD_ANG2	integer / two integers	Number of points for β -grid / grids	1,2,3,4 / 5,6,7,8,9,0
GRD_ANG3	integer / two integers	Number of points for γ -grid / grids	3,4 / 5,6,7,8,9,0
VGRID_FILE	default is “NO”	If “YES”, PES values at grid points are stored to/read from file	6,7,8,9,0
READ_EXPANSION	default is “NO”	If “YES”, the PES is represented by expansion over basis	except 2, 6
NMB_TERMS	Integer	Number of PES expansion terms	except 2, 6
TERMS	sets of integers	List of the expansion terms (labeled appropriately)	except 2, 6
CALC_EXPANSION	default is “NO”	If “YES”, the expansion coefficients are computed.	except 2, 6
IR_BGN, IR_FIN	Integer	The range of R -grid points used, defaults are 1 to GRD_R	all
RGRID_EQDS	default is “YES”	If “NO”, non-equidistant R -grid is generated by the code	all
RGRID_FILE	default is “NO”	If “YES”, user-defined R -grid is read from file	all
L_MAX	Integer	Maximum value of index λ in the expansion of the PES	all
M_MAX	Integer	Maximum value of index μ in the expansion of the PES	3,4
L1_MIN, L1_MAX	Integer	Min. and max. values of index λ_1 in the expansion of the PES	5,7,8,9,0
L2_MIN, L2_MAX	Integer	Min. and max. values of index λ_2 in the expansion of the PES	5,7,8,9,0
M1_MAX	Integer	Maximum value of index μ_1 in the expansion of the PES	5,7,8
M2_MAX	Integer	Maximum value of index μ_2 in the expansion of the PES	9,0
ODD_L1L2L	default is “NO”	Terms with odd values of $\lambda_1+\lambda_2+\lambda$ would be included	5,7,8,9,0
AXL_SYM1	integer, default is 1	Only μ_1 that are multiples of AXL_SYM1 are included	3,4,7,8,9,0
AXL_SYM2	integer, default is 1	Only μ_2 that are multiples of AXL_SYM2 are included	7,8,9,0
EQU_SYM1	integer, default is 1	Only λ_1 that are multiples of EQU_SYM1 are included	1,3,4,5,7,8,9,0
EQU_SYM2	integer, default is 1	Only λ_2 that are multiples of EQU_SYM2 are included	5,7,8,9,0
IDENTICAL_PES	default is “NO”	If “Yes”, the number of expansion terms is reduced by symmetry	5,0

PRINT_DIAGONAL	default is “NO”	If “YES”, prints diagonal elements of transition matrix	all
PARALLEL_IO	default is “NO”	If “YES”, read and write matrix elements in parallel	all
EQ_ANG_GRID	default is “YES”	Equidistant grids are used for α and γ (if “No”, Gauss-Legendre)	3,4,5,6,7,8,9,0
MTRX_PATH	default is “NO”	If “YES”, the matrix file/s is/are read from location indicated in the file named “Matrix_Path.DAT”	all
W3J_RECURSIVE	default is “NO”	If “YES” a recursive algorithm is used to compute Wigner 3j-symbols (instead of a simple algorithm). Works better for larger j .	all
MIJ_CUTOFF	real, default is 1.d-12 cm ⁻¹	Neglects matrix elements that are below this value	all
RETRUNCATE	default is “NO”	If “YES”, the matrix is re-truncated/rewritten using MIJ_CUTOFF	all
CUTOFF_R1	real, default is 6.d0 Bohr	Indicates first value of R for matrix analysis and truncation	all
CUTOFF_R2	real, default is 8.d0 Bohr	Indicates second value of R for matrix analysis and truncation	all
CALC_RMS	default is “NO”	If “YES”, RMS of PES expansion is computed for one value of R	1,3,4,5,7,8,9,0
RMS_GRD1	integer, default is 20	Sparse grid for RMS calculations, in α	5,7,8,9,0
RMS_GRD2	integer, default is 10	Sparse grid for RMS calculations, in β	1,3,4,5,7,8,9,0
RMS_GRD3	integer, default is 20	Sparse grid for RMS calculations, in γ	3,4,5,7,8,9,0
RMS_R	real, default is 8.d0 Bohr	Indicates the value of R for RMS calculations	1,3,4,5,7,8,9,0

III. Different Types of MQCT Calculations

By default, the most general full-coupled version of MQCT calculations is carried out by the code, in which the transitions due to the Coriolis coupling are included. Such “coupled-channel” calculations are referred to as CC-MQCT. One important option, initiated by the keyword `CS_APPROX=YES`, is to run the so-called *coupled-states* calculations, CS-MQCT [Semenov 2014], where the Coriolis-driven transitions are neglected. Within each j channel, the Coriolis term couples $2j + 1$ degenerate states labelled by m . In CS-MQCT, these Coriolis couplings and corresponding transitions are neglected, so, the calculations are done *independently* for various fixed values of m . Computational speed up by an order of magnitude is typical, due to simpler equations of motion and smaller number of coupled states.

New option in MQCT 2023 is adiabatic trajectory approximation, named AT-MQCT [Mandal 2020]. In this case, two runs of the code are required: During the first “adiabatic” run user can include only one state in the basis, the initial state, and indicate the keyword `SAVE_TRAJECT=YES`. The code propagates MQCT trajectories in this trivial basis (including the Coriolis driven transitions between m states of the initial channel) and saves all trajectory information to files. During the second run, with full basis set specified and with keyword `AT_APPROX=YES` indicated, the code propagates only the quantum equations of motion (to determine inelastic transition probabilities) reading pre-computed trajectory information from files. In this way, the coupling between classical and quantum parts of the system is neglected. Moreover, the evolution of potential energy along the trajectory recorded during the first run, can be used to automatically adjust the propagation time step for the second run, using a tolerance parameter, e.g., `AT_ADAPTOL=0.001` (with numerical value determined by convergence studies). In this case the minimum time step will be enforced using the value indicated by `TIME_STEP` keyword, and time steps will be increased along the trajectory to maximize the performance. Computational speed up by two orders of magnitude is typical (RK4 integrator is used in this case). In this version of the code, the CS-MQCT approximation can be used simultaneously with AT-MQCT approximation.

For AT-MQCT calculations the directory named `AT_APPROX_TRAJS` is created, with three more levels of sub-directories inside, that correspond to different collision energies and different degenerate states j and m . Information on each individual trajectory is saved to an unformatted file `AT_j_m_l_p.DAT`, where l is the orbital angular quantum number and p is exchange parity of two identical collision partners, zero or one (for non-identical partners this value is set to 0). If the Monte-Carlo sampling of initial conditions is used, the trajectories with different values of j and m are saved to the same directory, since these values are sampled randomly.

IV. Calculation of Potential Coupling Matrix for State-to-State Transitions

Within MQCT 2023 code there are two strategies/options for computing the potential coupling matrix, different by the method of PES representation and by data handling. Description of the format for these two options can be found in the file `user_suppl_pot.f` in the directory `PES_USER`, and the file `pes_sys_type.f` in the main code directory `MQCT_2023`

Option 1: is the default, equivalent to the keyword `EXPANSION=NO`. In this case the user should provide a subroutine `USER_DEFINED_PES` that generates the value of potential energy as a function of the molecule-molecule distance R and the internal coordinates (Euler angles in the body-fixed reference frame, and bond lengths). MQCT code will use this subroutine to compute elements of the state-to-state transition matrix directly, by numerical integration over the internal molecular degrees of freedom. Such calculations are done for every grid point of the molecule-molecule distance R and the data are stored in memory. For calculations of the collision process, when the values of matrix elements and their derivatives (for classical equations of motion) are needed at certain values of R along the trajectory, one-dimensional cubic spline of each matrix element is computed.

Option 2: keywords `READ_EXPANSION=YES`. In this case, the user is required to supply a data file `PES_EXPAN_TERMS.DAT` that contains pre-computed coefficients for analytical expansion of the PES (listed in Table 8 for different system types) at every grid point of the molecule-molecule distance R . The format of the data file is described in the subroutine `EXPAND_PES` (see Sec. III). Using these coefficients, MQCT will calculate elements of the state-to-state transition matrix analytically (using equations of Table 9) at every grid point of the molecule-molecule distance R , store them in the memory and finally spline (during the dynamics calculations, just like in the Option 1). This is convenient when these data are already available, say from literature. Importantly, the user can pre-run the MQCT code with the optional keyword `CALC_EXPANSION=YES` to generate the file `PES_EXPAN_TERMS.DAT`, as shown in several example files provided with the code, and is discussed in Sec. V.

By default, the reference orientation of the molecule corresponds to the symmetry plane coinciding with XZ-plane, in which case all matrix elements are real. Chiral molecules without any plane of symmetry cannot be treated with this version of the code. The optional keyword `REF_XZ=NO` can be used if the plane of symmetry is set to either YZ-plane or XY-plane. In these cases, matrix elements for ortho-para transitions become purely imaginary. If `REF_XZ=NO` is indicated, these imaginary matrix elements are saved in the same array as if they would be real, but an additional phase factor is introduced into the equation of motion to take this effect into account during the propagation of trajectories. At present time, this feature is available only for `SYS_TYPE=4` within Option 1 above.

Both the direct calculation of transition matrix (default, Option 1) and the projection of PES onto the basis set of analytic functions (Option 2, keyword `CALC_EXPANSION=YES`) deal with numerical integration over the internal degrees of freedom using multi-dimensional numerical quadrature. For both cases the number of quadrature points should be indicated by the user in the input file using keywords `GRD_VIB`, `GRD_ANG1`, `GRD_ANG2`, and `GRD_ANG3`. Integration over vibrational coordinate uses an equidistant grid quadrature for automatically generated Morse oscillator functions, or optionally a user-specified non-equidistant grid with the weights

indicated for each point in the file `USER_DEFINED_BASIS.DAT`. Gauss-Legendre quadrature is used for angle β only (`GRD_ANG2`). Integrals over α and γ are computed using an equidistant grid by default, but Gauss-Legendre quadrature can also be requested using a keyword `EQ_ANG_GRID=NO`. For all molecule + atom systems (`SYS_TYPE=1` to `4`) only one integer value should be assigned to each of `GRD_VIB`, `GRD_ANG1`, `GRD_ANG2`, and `GRD_ANG3`. But for all molecule + molecule systems (`SYS_TYPE=5` to `0`) two integer numbers should be given sequentially, separated by coma. Note, however, that depending on the system type, some of these numbers are dummy (not used) and can be arbitrary. Details are given below:

`SYS_TYPE=1`: for diatomic + atom integration is carried out along β -angle only, with the number of points `GRD_ANG2`. The values of `GRD_ANG1` and `GRD_ANG3` are dummy. Vibrational degree of freedom is dummy for all `SYS_TYPE` but `2` and `6`.

`SYS_TYPE=2`: for vibrating diatomic + atom, besides β -angle described above, the integral includes `GRD_VIB` points for vibration. The values of `GRD_ANG1` and `GRD_ANG3` are dummy, just as in `SYS_TYPE=1`.

`SYS_TYPE=3` and `4`: for any top + atom the number of points for angles β and γ is indicated by `GRD_ANG2` and `GRD_ANG3`, respectively. The value of `GRD_ANG1` is dummy.

`SYS_TYPE=5`: for diatom + diatom the number of points along two β -angles is given by two entries of the keyword `GRD_ANG2`, while the number of points for α -angle is indicated by the *second* entry of the keyword `GRD_ANG1`. The first entry of the keyword `GRD_ANG1`, and both entries of the keyword `GRD_ANG3` are dummy.

`SYS_TYPE=6`: for vibrating diatom + diatom the number of points along each bond length is indicated by two entries of the keyword `GRD_VIB`, in addition to the angular coordinates of `SYS_TYPE=5`.

`SYS_TYPE=7` and `8`: for any top + diatom the number of points along two β -angles is given by two entries of the keyword `GRD_ANG2`, the number of points along α is given by the second entry of `GRD_ANG1` (the first entry is dummy), while the number of points along γ is given by the first entry of the keyword `GRD_ANG3` (the second entry is dummy).

`SYS_TYPE=9` and `0`: for top + top systems the number of points along α is given by the second entry of `GRD_ANG1` (only the first entry is dummy), the number of points along two β -angles is given by two entries of the keyword `GRD_ANG2`, the number of points along two γ -angles is given by two entries of the keyword `GRD_ANG3`.

V. Compiling and Running the Code, and Choosing Efficient Parallelization Strategies

Compilation of the PES function and of MQCT code is done by using a `Makefile`. Our version of `Makefile` is provided as an example in the main directory. Users are expected to change variables in the `Makefile` as needed for their computer architecture, directory structure and PES function. The example `Makefile` contains a brief description at the top of the file to describe how it should work. Users would make changes in the section that follows. First, one needs to provide a path to PES subroutines as variable named `pes_dir`. Next, in the variable named `pes_files`, one would write the name of the Fortran file (or files separated by a blank space) that contains the PES subroutine, with the extension `*.o` (i.e., all object files of the PES routine). After that, users should provide their choice of name for the executable file as variable `exe_name`. Finally, if the MQCT codes are not present in the same directory (as shown in the example `Makefile`), then one would have to provide a path to the MQCT code files as variable `src_dir`.

Next step in modifying the `Makefile` is to provide appropriate names for the compiler and linear algebra libraries. For that, the user should indicate the name of the Fortran compiler with MPI option in the variable `FC`. Then any flags for the Fortran compiler should be mentioned in the variable `FFLAGS`, such as optimization level, debugging or traceback. At the end, one should indicate the name of the linear algebra libraries in the variable `LDFLAGS`, such as `LAPACK`, `BLAS` and `MKL`.

After making and saving all changes in the `Makefile`, and before proceeding to the compilation stage, users should make sure that all the necessary modules for Fortran compiler and math libraries are loaded. It is also recommended that prior to compilation users clean their old files by using `$make clean` command. After that, simply use `$make` to compile everything. If the compilation is successful, the directory `./bin` will contain an executable file and it is recommended to use this file in further calculations.

In the example directory, our `Makefile` is prepared for $\text{H}_2\text{O} + \text{H}_2\text{O}$ system. The PES of the molecular system is located in the directory `./PES_DIRECTORIES/PES_H2OH2O` which is assigned to the variable `pes_dir`. For this system, all subroutines to evaluate the PES are combined in one FORTRAN file `PES_H2OH2O.f`. Therefore, the variable `pes_files` is assigned only one filename `PES_H2OH2O.o`. The name of the executable as indicated in the variable `exe_name` is `mqct_H2OH2O`. And finally, the MQCT codes are located in the directory named `MQCT_SOURCE_CODES` in the root, which is assigned to `src_dir`.

The input file for MQCT should have the extension `*.inp`, and its name should be placed in the file `INPUT_NAME.inp`. This permits the user to store multiple input files (e.g., for different molecules) in the program directory but run actual calculations with one specific input file.

There are two levels of parallelization in the code. At the first level, propagation of each trajectory can be done by multiple processors used as a group (e.g., all processors of a node) to compute right-hand sides of the classical and quantum equations of motion. This requires some minimal message passing. At the second level, propagation of different trajectories can be assigned to different groups, which requires virtually no message passing. The program attempts to evenly split all requested trajectories between the groups.

For example, if the code is submitted for execution using 60 nodes of the machine with 32 cores per node (1920 processors total) the following option, indicated in the \$SYSTEM block:

```
MPI_PERTRAJ=32, NMB_TRAJ=300
```

will result in formation of 60 groups, and assignment of 5 trajectories per group. Typically, trajectories with larger impact parameters are shorter and faster to propagate. Thus, for an equalized load, and optimal use of resources, it is recommended to assign several trajectories per group, not just one. Note that NMB_TRAJ is an optional keyword used only in conjunction with Monte-Carlo sampling of the initial states (see Sec. VIII below). In regular (not a Monte-Carlo) calculations, the code automatically determines the required number of MQCT trajectories and attempts to split them evenly between the groups.

Several practical aspects of MQCT calculations should be outlined. In principle, the calculations of potential coupling matrix and the subsequent calculations of collision trajectories can be done in a single run of MQCT, but this is practical for simple systems only, when the CPU effort and the memory requirement are relatively low. Namely, if the keyword PROG_RUN=YES is indicated, while the matrix is not yet available, the code will compute the matrix by direct integration (Option 1, the default) and will proceed to the propagation of trajectories, all in one run. Calculation of different matrix elements is split about equally between different processors and there is not any constraint on the number of processors at the matrix stage. When the code proceeds to the propagation of the equations of motion, it allocates a group of processors (MPI_PERTRAJ) for calculation of the right-hand-side of each equation of motion. Within a group, each processor holds in its memory only a part of the matrix, and the overall cost of trajectory propagation is split about equally. If the total number of processors in the run is N_PROC, the code will start propagating a set of $N_PROC/MPI_PERTRAJ$ trajectories, which should be an integer. When a group of processors finishes one trajectory, it takes the next, and so on. Therefore, the number of processors is limited by $N_PROC \leq MPI_PERTRAJ * NMB_TRAJ$. If the number of processors is larger than this, the code will stop with an error message. If SAVE_MTRX=YES is indicated, then, prior to the propagation of trajectories, the code will save truncated matrix to a single file, unformatted by default (MTRX_UF.DAT) or formatted (MTRX.DAT) if the keyword UNFORMAT=NO is specified. This procedure works but has a bottleneck: Since all pieces of the matrix computed by different processors are sent to one processor, a very large memory per processor is required (in order to store the entire matrix). Writing the entire matrix to one file by one processor is also slow and inefficient (since other processors are idling). Therefore, for calculations with a large matrix, a multistep procedure with parallel-IO is recommended, as explained below.

For complicated molecular systems with many states and large state-to-state transition matrix it is recommended that the matrix is pre-computed in a separate run of MQCT following one parallelization strategy, and only after that the trajectories are propagated using a different number of processors and different parallelization strategy. Namely, for the “matrix” run two keywords SAVE_MTRX=YES, PROG_RUN=NO should be indicated, and the number of processors should be as large as possible. The overhead

of parallelization is negligible for matrix calculations, and this approach permits to reduce the wall-clock time. The subsequent “propagation” run is initiated by two keywords: `READ_MTRX=YES`, `PROG_RUN=YES`. In order to reduce message passing between the nodes, one may want to fit a whole number of processor groups in each node (for example, ten groups per node), by choosing an appropriate group-size `MPI_PERTRAJ`. However, for very large systems the memory requirement may become more important. Remember that each group of processors will have to hold the entire matrix in the memory. Therefore, expanding the group onto several nodes may be unavoidable for machine architectures with small memory.

While the matrix calculations *per se* do not require large memory, the matrix collection and redistribution by one processor, as well as saving to (and reading from) single file represent a bottleneck. To avoid this problem, users can indicate the keyword `PARALLEL_IO=YES` during both “matrix” and “propagation” runs. At the “matrix” stage, this option tells individual processors to save their pieces of computed (and truncated) matrix into individual output files without any message passing. The number of files will be equal to the number of processors used for this run. During the subsequent “propagation” run, with `PARALLEL_IO=YES` indicated, the elements of truncated matrix will be distributed evenly between processors of each group, but each processor will read and use only its assigned chunk of the matrix. This procedure avoids reading the entire matrix by one processor or holding the entire matrix in the memory of one processor.

While two separate runs are optional when the matrix is calculated by direct integration (Option 1), they are mandatory in the case when the PES expansion is computed (prior to using Option 2). Namely, if Option 2 is used, then for the first run, initiated by the keyword `CALC_EXPANSION=YES` to generate the PES expansion coefficients, the number of processors must be equal to the number of requested expansion terms, since each processor will be responsible for computing one expansion coefficient. The code will compute the expansion coefficients and will stop, without proceeding to the calculations of matrix elements or trajectories. Then, user should replace `CALC_EXPANSION` in the input file by `READ_EXPANSION=YES` and run the code again with the number of processors appropriate for calculations of matrix elements (keywords `SAVE_MTRX=YES`, `PROG_RUN=NO`). Finally, the code should be run a third time for the propagation of trajectories (keywords `READ_MTRX=YES`, `PROG_RUN=YES`), with an appropriate value of `MPI_PERTRAJ` set up. The keyword `PARALLEL_IO=YES` can be used during the “matrix” and “trajectories” calculations, as explained above.

In this procedure, one should be careful about the units of distance and energy. The file with expansion terms `PES_EXPAN_TERMS.DAT`, printed by the code, will always contain distance in Bohr and energy in the same units that are used by the potential energy surface routine (employed for the calculations of expansion). If the expansion is used further (to compute the state-to-state transition matrix and/or the collision dynamics), the units of distance should be set as Bohr and the units of energy should stay the same as in the potential energy surface routine. However, the matrix elements computed by the code will always be saved in atomic units (Bohr for distance and Hartree for energy).

If the angular grid is very large, and, in particular, when the number of R -grid points is large, it may be convenient to split calculations of the PES expansion into several runs. To do that, user can specify the range of R -grid to cover in one run, using keywords `IR_BGN` and `IR_FIN`. Results of successive runs are combined automatically into a single file for the expansion terms and coefficients.

When computing the state-to-state transition matrix by direct integration (Option 1), or computing the PES expansion by projection (Option 2), it may be advantageous, in terms of CPU time, to keep in the memory the values of potential at the grid points, rather than calling the PES subroutine each time when the value for new point is needed. However, a very large grid (for larger molecules and complicated PES) may not fit as a whole into the memory of one CPU. For this case the option `VGRID_FILE=YES` is recommended. The code first generates the PES at the points of the grid and saves these data to the unformatted file `VGRID_UF.DAT`. The number of processors should be at least equal to the number of points of R -grid, or larger. Then the code loads this information into the memory of processors to compute matrix elements (or the expansion coefficients) but does it by slices, sequentially for each value of R on the grid, since the calculation at each value of R is independent. Different processors will be responsible for computing different elements of the matrix, or different expansion terms. Note that if the expansion is computed, the code will save the data file `VGRID_UF.DAT` and will normally stop (except a rare special case when the number of R -grid points is equal to the number of the PES expansion terms). It should be run again with the number of CPUs equal to the number of the expansion terms. The code will read the data file `VGRID_UF.DAT` and proceed to the calculations of the expansion coefficients, one term per processor. Calculations of matrix elements and the propagation of trajectories should be done in the following independent runs, as explained above.

State-to-state transition matrix elements computed and used by the code are labeled by one index that runs through all matrix elements (all transitions). Information about two states coupled by given matrix element is contained in a separate array and is stored in the same matrix file. If the keyword `PARALLEL_IO=YES` is used and the matrix is saved to multiple files, the range of matrix elements saved to each file is embedded in the file name using the indexes of first and last matrix elements in this file, `Mij_xxx_yyy.DAT`. Inside the file, we save sequentially the index of transition, the value of R -grid, and the value of matrix element in Hartree, running the inner cycle over R -grid and the outer cycle over the index of transition. Information about two states coupled by given matrix element is contained in a separate index file, `Ind_xxx_yyy.DAT`. Directory `MATRIX_TRUNCATED` holds matrix files created by different processors. In order to take full advantage of symmetry (when the symmetry is not obvious) the code automatically disregards transitions described by matrix elements with absolute values smaller than `MIJ_CUTOFF=1.d-12` (in the units of cm^{-1}). These matrix elements are neglected, and the corresponding transitions are removed from the list. If needed, an alternative value of the cut-off can be specified, for example, to make calculations more affordable by disregarding transitions between the weakly coupled states. This truncation scheme represents simple but very efficient approximation. A suitable value of `MIJ_CUTOFF` should be determined by convergence studies. Two values of R distance (where the matrix elements are analyzed for this truncation) can be indicated using keywords `CUTOFF_R1`

and `CUTOFF_R2` (the default values are 6 and 8 Bohr). For each transition, the code takes the values of matrix element at two grid points closest to `CUTOFF_R1` and `CUTOFF_R2` and compares those versus `MIJ_CUTOFF`. If both are smaller, this transition is disregarded to create a truncated (reduced) matrix. One can easily re-truncate the existing matrix, by moving it to the directory `MATRIX_FILES`, indicating the value of desired `MIJ_CUTOFF` and running the code one more time (with the same number of processors that was used for calculations of the original matrix) using keywords `READ_MTRX=YES`, `RETRUNCATE=YES`, `PROG_RUN=NO`. New directory `MATRIX_TRUNCATED` will be created with new files that will hold the re-truncated matrix. The new value of `MIJ_CUTOFF` should be larger than the old one. If one wants a matrix with smaller cut-off value (a larger matrix), it has to be recomputed from scratch.

Note that `MIJ_CUTOFF` is implemented differently in the serial version of IO, appropriate for simple systems and smaller matrixes. The code still compares matrix elements at two points of R -grid closest to `CUTOFF_R1` and `CUTOFF_R2` versus `MIJ_CUTOFF`, but if both are smaller, the value of zero matrix element is written into the `MTRX.DAT` file at all values of R -grid (the transition is effectively neglected, but the matrix size remains the same).

VI. PES Interface and Other User-Supplied Subroutines and Data Files

Formally, the potential energy surface subroutine `USER_DEFINED_PES` operates with the same coordinates for all system types, but some of these coordinates are dummy variables. Namely, the input for PES subroutine requires, besides the molecule-molecule distance R , one vibrational coordinate and three Euler angles for each collision partner (see the file `user_suppl_pot.f` in the directory `PES_USER`, or the file `pes_sys_type.f` in the code directory `MQCT_2026`). However, the vibrational coordinate is used only for `SYS_TYPE=2, 6` where the vibrational motion of the diatomic is explicitly treated, while it is a dummy variable for all other system types. In the future, our plan is to add one vibrational degree of freedom (such as bending motion in triatomic molecules) for other system types, but this is not yet implemented in the present release of the code. The subroutine `USER_DEFINED_PES` can also be used as a “layer” between the external PES subroutine and `MQCT`, for example, to convert degrees into radians, or to transform spherical angles into Euler angles. Note that a subroutine `MFTOBF_CONV` (in the file `user_suppl_pot.f`) is available for conversion of the PES from a popular space-fixed reference frame to the body-fixed reference frame used by `MQCT` for `SYS_TYPE=7, 8`.

Concerning the angular coordinates, some of them are dummy as described in Sec. II above, and some of them are arbitrary (fixed at zero value here). For completeness, we summarized these properties in Table 6, where each dummy variable is indicated by a dash. We want to emphasize one more time that our reference frame, called the body-fixed reference frame tied to the molecule-molecule

vector, is different from coordinates used in some other codes (such as MOLSCAT) where the reference frame is tied to one of the molecules. The words “azimuthal” and “polar” are given in the cases of diatomics, to underline the meaning of those rotations.

Table 6: Degrees of freedom in the user-supplied PES subroutine

SYS_TYPE	R	r_1	r_2	α_1	β_1	γ_1	α_2	β_2	γ_2
1	distance	--	--	--	Polar	zero	--	--	--
2	distance	vibration	--	--	Polar	zero	--	--	--
3	distance	--	--	zero	Euler	Euler	--	--	--
4	distance	--	--	zero	Euler	Euler	--	--	--
5	distance	--	--	--	Polar	zero	Azimuthal	polar	--
6	distance	vibration	vibration	--	Polar	zero	Azimuthal	polar	--
7	distance	--	--	zero	Euler	Euler	Azimuthal	polar	--
8	distance	--	--	zero	Euler	Euler	Azimuthal	polar	--
9	distance	--	--	zero	Euler	Euler	Euler	Euler	Euler
0	distance	--	--	zero	Euler	Euler	Euler	Euler	Euler

It is recommended that users print and inspect 2D-slices of the PES connected to MQCT program. This can be done using the optional keyword PRN_PES=YES alongside the default OPTION 1 for computing state-to-state matrix elements (the size of rotational basis set is irrelevant, can be small). Two degrees of freedom can be varied, others must be fixed at certain values, dummy variables are ignored but all nine variables must be indicated in the input of the keyword FIX_VAR (distances in the units of connected PES, angles in degrees). The output 2D-slice will use the units of the connected PES for energy and distances, and degrees for angles.

For several special cases listed below, some of the input data must be generated externally by users and supplied in separate files. The following table gives a brief description of these data files (extension *.DAT). Examples of these files can be found in the directory EXAMPLES_IO. In order to create properly formatted files, users can employ our subroutines supplied with the code, all located in the TOOLBOX_CODES. They are listed in the table below. Note that these are not ready-to-use utility programs to generate these data, but merely the examples of data formats required by MQCT code.

Table 7: Description of the input data files for MQCT calculations

Data File Name	Generating Subroutine	Brief Description
PES_EXPAN_TERMS.DAT	EXPAND_PES	Contains R -dependence of the expansion coefficients for analytic representation of the PES (as in Table 8 below). Works for all SYS_TYPE, except 2 and 6 where the vibrational motion is involved.
USER_DEFINED_BASIS.DAT	DEFINE_BASIS See example file	Contains channel labels (quantum numbers), energies, and wave functions of the externally computed ro-vibrational states. For SYS_TYPE=2 and 6 the vibrational wave functions should be pre-computed on a grid, with indicated grid points and weights. For SYS_TYPE=4, 8, 9, 0 the rotational states (<i>e.g.</i> , of Kyrø Hamiltonian) should be represented by expansion over the basis set of symmetric-top eigenstates. Depending on system type, this includes one or both collision partners.
USER_DEFINED_RGRID.DAT	See example file	Contains user defined (<i>e.g.</i> , non-equidistant) grid for the molecule-molecule distance R . It can be useful for large amplitude vibrational motion states. Must be in the units of Bohr.
STATES_TO_EXCLUDE.DAT	See example file	Optional. Undesired states can be excluded by listing in this file the state numbers (as they appear in the file STATES.out), and setting the keyword EXCLUDE_STATES=YES.

Note that the default method of describing ro-vibrational states is expected to be reasonably accurate only for low to moderate rotational and vibrational excitations, since it neglects ro-vibrational interaction. A more reliable approach, accurate up to dissociation limit, is to compute numerically accurate ro-vibrational states using an external code (not provided) and feed them as input for the MQCT code for one (SYS_TYPE=2) or both (SYS_TYPE=6) collision partners. This is achieved by the keyword LEVELS_FILE=YES. If specified, the file named USER_DEFINED_BASIS.DAT is also required (copied or linked to the code directory), that should contain energies and wave functions of pre-computed states. The number of states in the files should be equal to NMB_CHNLS (indicated in the block \$BASIS). The values and weights of grid points for numerical integration (can be non-equidistant) should also be specified. The number of points should be equal to GRD_VIB (indicated in the block \$POTENTIAL). Examples of such files are distributed with the code.

VII. Expansion of PES Over the Basis Set of Analytic Functions

In MQCT the PES can be expanded over the basis set of analytic functions specific to the system type (by indicating keyword `CALC_EXPANSION=YES`). Those are listed in Table 8, along with the labels of such expansion terms. Note that all our expansions use three angles of active Euler rotations for orientation of each molecule relative to the molecule-molecule axis z , which is called body-fixed reference frame (which is different from a popular set-up where one of the molecules is fixed in the lab, while the other is moved relative to it and is rotated relative to the lab axes). To define the expansion terms by the keyword `TERMS`, user will specify a set of “quantum numbers” for each term, separated by coma. The order of indicated expansion terms defines the order in which they will be handled (computed by subroutines `USER_DEFINED_TERMS` or `CALC_EXPANSION=YES`, written and read from file `PES_EXPAN_TERMS.DAT`, summed into the matrix element, *etc.*). The total number of terms to be employed must be specified by the keyword `NMB_TERMS`. If user does not wish to list all terms, the code can automatically assign them based on the keywords `L1_MAX`, `M1_MAX`, `L2_MAX`, `M2_MAX` and `L_MAX`, as appropriate to each system type (see Table 8). In this case, the code will generate all the terms and save them into a file named `EXPANSION_TERMS_LIST.DAT` and will stop. The user will then need to incorporate these terms within the input (copy/paste). One can add/eliminate some terms by hand at this point if one wants to do that.

The ranges of “quantum numbers” for expansion terms are:

$$\lambda_1, \lambda_2, \lambda \geq 0; \quad |\lambda_1 - \lambda_2| \leq \lambda \leq \lambda_1 + \lambda_2; \quad 0 \leq \mu_1 \leq \lambda_1 \text{ (positive only); } \quad -\lambda_2 \leq \mu_2 \leq \lambda_2 \text{ (positive if } \mu_1 = 0).$$

Note that in the expressions of Table 8 two terms in each square bracket account for the invariance of potential with respect to reflection through origin, using phase factor $(-1)^{\lambda_1 + \mu_1 + \lambda_2 + \mu_2 + \lambda}$. Moreover, for the PESs of two identical molecules (`SYS_TYPE=0, 5`), indicated by optional keyword `IDENTICAL_PES=YES`, the expansion functions are further symmetrized with respect to exchange of two molecules, by combining the expansion terms as indicated in Table 8 with phase factor $(-1)^{\lambda_1 + \lambda_2}$, and, simultaneously the number of terms is reduced by two additional conditions: $\lambda_2 \leq \lambda_1$, and, if $\lambda_2 = \lambda_1$ then $|\mu_2| \leq \mu_1$.

Also note that for linear molecules only the terms with even values of $\lambda_1 + \lambda_2 + \lambda$ are required. In the case of non-linear molecules, we found that the terms with odd values of $\lambda_1 + \lambda_2 + \lambda$ are also small and therefore can often be neglected (although they are not strictly zero and may play role in some cases). The keyword `ODD_L1L2L=YES` may be used to generate the terms with odd values of $\lambda_1 + \lambda_2 + \lambda$.

The expansion functions τ in Table 8 are mutually orthogonal, are complex in general, but are normalized only in those cases when the square brackets are not present (although, each individual term in the square bracket is also normalized). In the code, for simplicity and efficiency, the expansion coefficients are obtained by projection of potential only onto the first term in each square bracket: $v(R) = \langle \text{first term of } \tau(\Lambda) | V(R, \Lambda) \rangle$, and then this $v(R)$ is used for every term in the bracket). Renormalization of the composite function is not needed because each term is used as a separate normalized basis function. These expansion coefficients are always real. They are used to construct elements of state-to-state transition matrix using analytical expressions summarized in Table 9.

Table 8: PES expansion functions in MQCT calculations

System Type	System Description	Labels of Expansion Terms	Expansion Functions
1	rigid diatom + atom	λ	$\tau_\lambda(\beta) = Y_0^\lambda(\beta, 0)$
2	vibr.diatom + atom		
3	sym. top + atom	λ, μ	$\tau_{\lambda\mu}(\beta, \gamma) = \frac{1}{1 + \delta_{\mu 0}} [(-1)^\mu Y_{+\mu}^\lambda(\beta, \gamma) + Y_{-\mu}^\lambda(\beta, \gamma)]$
4	asym. top + atom		
5	rigid diatom + rigid diatom	$\lambda_1, \lambda_2, \lambda$	$\tau_{\lambda_1 \lambda_2 \lambda}(\beta_1, \alpha_2, \beta_2) = \sum_{\eta = -\min(\lambda_1, \lambda_2)}^{+\min(\lambda_1, \lambda_2)} C_{\lambda_1, \eta, \lambda_2, -\eta}^{\lambda, 0} Y_{+\eta}^{\lambda_1}(\beta_1, 0) Y_{-\eta}^{\lambda_2}(\beta_2, \alpha_2)$ If IDENTICAL_PES=YES, combine with: $+ C_{\lambda_2, \eta, \lambda_1, -\eta}^{\lambda, 0} Y_{+\eta}^{\lambda_2}(\beta_1, 0) Y_{-\eta}^{\lambda_1}(\beta_2, \alpha_2)$
6	vibr. diatom + vibr. diatom		
7	sym. top + rigid diatom	$\lambda_1, \mu_1, \lambda_2, \lambda$	$\tau_{\lambda_1 \mu_1 \lambda_2 \lambda}(\beta_1, \gamma_1, \alpha_2, \beta_2) = \frac{1}{1 + \delta_{\mu_1 0}} \sqrt{\frac{2\lambda_1 + 1}{4\pi}} \sum_{\eta = -\min(\lambda_1, \lambda_2)}^{+\min(\lambda_1, \lambda_2)} C_{\lambda_1, \eta, \lambda_2, -\eta}^{\lambda, 0}$ $\times [D_{\eta, +\mu_1}^{\lambda_1*}(0, \beta_1, \gamma_1) + (-1)^{\lambda_1 + \mu_1 + \lambda_2 + \lambda} D_{\eta, -\mu_1}^{\lambda_1*}(0, \beta_1, \gamma_1)] \times Y_{-\eta}^{\lambda_2}(\beta_2, \alpha_2)$
8	asym. top + rigid diatom		
9	sym. top + asym. top	$\lambda_1, \mu_1 \lambda_2, \mu_2, \lambda$	$\tau_{\lambda_1 \mu_1 \lambda_2 \mu_2 \lambda}(\beta_1, \gamma_1, \alpha_2, \beta_2, \gamma_2)$ $= \frac{1}{1 + \delta_{\mu_1 0} \delta_{\mu_2 0}} \sqrt{\frac{2\lambda_1 + 1}{8\pi^2}} \sqrt{\frac{2\lambda_2 + 1}{8\pi^2}} \sum_{\eta = -\min(\lambda_1, \lambda_2)}^{+\min(\lambda_1, \lambda_2)} C_{\lambda_1, \eta, \lambda_2, -\eta}^{\lambda, 0}$ $[D_{+\eta, +\mu_1}^{\lambda_1*}(0, \beta_1, \gamma_1) D_{-\eta, +\mu_2}^{\lambda_2*}(\alpha_2, \beta_2, \gamma_2)$ $+ (-1)^{\lambda_1 + \mu_1 + \lambda_2 + \mu_2 + \lambda} D_{+\eta, -\mu_1}^{\lambda_1*}(0, \beta_1, \gamma_1) D_{-\eta, -\mu_2}^{\lambda_2*}(\alpha_2, \beta_2, \gamma_2)]$ If IDENTICAL_PES=YES, combine with: $+ C_{\lambda_2, \eta, \lambda_1, -\eta}^{\lambda, 0} [(-1)^{\lambda_1 + \lambda_2} D_{+\eta, +\mu_2}^{\lambda_2*}(0, \beta_1, \gamma_1) D_{-\eta, +\mu_1}^{\lambda_1*}(\alpha_2, \beta_2, \gamma_2)$ $+ (-1)^{\mu_1 + \mu_2 + \lambda} D_{+\eta, -\mu_2}^{\lambda_2*}(0, \beta_1, \gamma_1) D_{-\eta, -\mu_1}^{\lambda_1*}(\alpha_2, \beta_2, \gamma_2)]$
0	asym. top + asym. top		

Note: The relationship $(-1)^m Y_m^j = \sqrt{(2j+1)/(4\pi)} D_{0m}^{j*}$ is used for system types 3 and 4. The relationship $Y_m^j = \sqrt{(2j+1)/(4\pi)} D_{m0}^{j*}$ is used for system types 5 to 8. Kronecker symbol is implemented in the code using two loops for $\mu > 0$ but only one loop for $\mu = 0$.

Table 9: Computation of state-to-state transition matrix elements using PES expansion

System Type	System Description	State-to-state transition matrix equations
1	rigid diatom + atom	$\langle j''m V(\Lambda) j'm\rangle = \sqrt{\frac{2j'+1}{2j''+1}} \sum_{\lambda} v_{\lambda}(R) \sqrt{\frac{2\lambda+1}{4\pi}} C_{j',m,\lambda,0}^{j'',m} C_{j',0,\lambda,0}^{j'',0}$
2	vibr. diatom + atom	$\langle j''mv'' V(\Lambda) j'mv'\rangle = \sqrt{\frac{2j'_1+1}{2j''_1+1}} \sum_i w_i \psi_{v'_1}(r_i) \psi_{v''_1}(r_i) \sum_{\lambda} v_{\lambda}^i(R) \sqrt{\frac{2\lambda+1}{4\pi}} C_{j',m,\lambda,0}^{j'',m} C_{j',0,\lambda,0}^{j'',0}$
3	sym. top + atom	$\langle j''mk''\varepsilon'' V(\Lambda) j'mk'\varepsilon'\rangle = \sqrt{\frac{2j'+1}{2j''+1}} \frac{1}{\sqrt{2(1+\delta_{k',0})2(1+\delta_{k'',0})}} \sum_{\lambda\mu} \frac{v_{\lambda\mu}(R)}{1+\delta_{\mu 0}} \sqrt{\frac{2\lambda+1}{4\pi}} C_{j',m,\lambda,0}^{j'',m}$ $\left[\left(C_{j',k',\lambda,\mu}^{j'',k''} + \varepsilon' C_{j',-k',\lambda,\mu}^{j'',k''} + \varepsilon'' C_{j',k',\lambda,\mu}^{j'',-k''} + \varepsilon' \varepsilon'' C_{j',-k',\lambda,\mu}^{j'',-k''} \right) + (-1)^{\mu} \left(C_{j',k',\lambda,-\mu}^{j'',k''} + \varepsilon' C_{j',-k',\lambda,-\mu}^{j'',k''} + \varepsilon'' C_{j',k',\lambda,-\mu}^{j'',-k''} + \varepsilon' \varepsilon'' C_{j',-k',\lambda,-\mu}^{j'',-k''} \right) \right]$
4	asym. top + atom	$\langle j''m V(\Lambda) j'm\rangle = \sqrt{\frac{2j'+1}{2j''+1}} \sum_{\lambda\mu} \frac{v_{\lambda\mu}(R)}{1+\delta_{\mu 0}} \sqrt{\frac{2\lambda+1}{4\pi}} C_{j',m,\lambda,0}^{j'',m} \sum_{k''=-j''}^{+j''} \sum_{k'=-j'}^{+j'} b_{k''} b_{k'} \left(C_{j',k',\lambda,\mu}^{j'',k''} + (-1)^{\mu} C_{j',k',\lambda,-\mu}^{j'',k''} \right)$
5	rigid diatom + rigid diatom	$\langle j''m j''_1 j''_2 V(\Lambda) j'm j'_1 j'_2 \rangle = \sqrt{\frac{2j''_1+1}{2j'_1+1}} \sqrt{\frac{2j''_2+1}{2j'_2+1}} \sum_{\lambda_1 \lambda_2 \lambda} v_{\lambda_1 \lambda_2 \lambda}(R) \sqrt{\frac{2\lambda_1+1}{4\pi}} \sqrt{\frac{2\lambda_2+1}{4\pi}}$ $\times C_{j''_1,0,\lambda_1,0}^{j'_1,0} C_{j''_2,0,\lambda_2,0}^{j'_2,0} \left\{ \sum_{m_1=-j'_1}^{+j'_1} C_{j'_1,m_1,j'_2,m-m_1}^{j',m} \left(\sum_{\eta=-\min(\lambda_1,\lambda_2)}^{+\min(\lambda_1,\lambda_2)} C_{\lambda_1,\eta,\lambda_2,-\eta}^{\lambda,0} C_{j''_1,m_1-\eta,j''_2,m-(m_1-\eta)}^{j'',m} C_{j''_1,m_1-\eta,\lambda_1,\eta}^{j'_1,m_1} C_{j''_2,m-(m_1-\eta),\lambda_2,-\eta}^{j'_2,m-m_1} \right) \right\}$ If IDENTICAL_PES=YES, combine with: $+ C_{j''_1,0,\lambda_2,0}^{j'_1,0} C_{j''_2,0,\lambda_1,0}^{j'_2,0} \left\{ \sum_{m_1=-j'_1}^{+j'_1} C_{j'_1,m_1,j'_2,m-m_1}^{j',m} \left(\sum_{\eta=-\min(\lambda_1,\lambda_2)}^{+\min(\lambda_1,\lambda_2)} C_{\lambda_2,\eta,\lambda_1,-\eta}^{\lambda,0} C_{j''_1,m_1-\eta,j''_2,m-(m_1-\eta)}^{j'',m} C_{j''_1,m_1-\eta,\lambda_2,\eta}^{j'_1,m_1} C_{j''_2,m-(m_1-\eta),\lambda_1,-\eta}^{j'_2,m-m_1} \right) \right\}$

6	vibr. diatom + vibr. diatom	$\langle j'' m j_1'' j_2'' v_1'' v_2'' V(\Lambda) j' m j_1' j_2' v_1' v_2' \rangle = \sqrt{\frac{2j_1''+1}{2j_1'+1}} \sqrt{\frac{2j_2''+1}{2j_2'+1}} \sum_{i_1} w_{i_1} \psi_{v_1'}(r_{i_1}) \psi_{v_1''}(r_{i_1}) \sum_{i_2} w_{i_2} \psi_{v_2'}(r_{i_2}) \psi_{v_2''}(r_{i_2}) \sum_{\lambda_1 \lambda_2 \lambda} v_{\lambda_1 \lambda_2 \lambda}^{i_1 i_2}(R) \sqrt{\frac{2\lambda_1+1}{4\pi}} \sqrt{\frac{2\lambda_2+1}{4\pi}}$ $\times C_{j_1'',0,\lambda_1,0}^{j_1',0} C_{j_2'',0,\lambda_2,0}^{j_2',0} \left\{ \sum_{m_1=-j_1'}^{+j_1'} C_{j_1',m_1,j_2',m-m_1}^{j_1',m} \left(\sum_{\eta=-\min(\lambda_1,\lambda_2)}^{+\min(\lambda_1,\lambda_2)} C_{\lambda_1,\eta,\lambda_2,-\eta}^{\lambda,0} C_{j_1'',m_1-\eta,j_2'',m-(m_1-\eta)}^{j_1'',m} C_{j_1',m_1-\eta,\lambda_1,\eta}^{j_1',m_1} C_{j_2',m-(m_1-\eta),\lambda_2,-\eta}^{j_2',m-m_1} \right) \right\}$ If IDENTICAL_PES=YES, combine with: $+ C_{j_1'',0,\lambda_2,0}^{j_1',0} C_{j_2'',0,\lambda_1,0}^{j_2',0} \left\{ \sum_{m_1=-j_1'}^{+j_1'} C_{j_1',m_1,j_2',m-m_1}^{j_1',m} \left(\sum_{\eta=-\min(\lambda_1,\lambda_2)}^{+\min(\lambda_1,\lambda_2)} C_{\lambda_2,\eta,\lambda_1,-\eta}^{\lambda,0} C_{j_1'',m_1-\eta,j_2'',m-(m_1-\eta)}^{j_1'',m} C_{j_1',m_1-\eta,\lambda_2,\eta}^{j_1',m_1} C_{j_2',m-(m_1-\eta),\lambda_1,-\eta}^{j_2',m-m_1} \right) \right\}$
7	sym. top + rigid diatom	$\langle j'' m j_1'' k'' \varepsilon'' j_2'' V(\Lambda) j' m j_1' k' \varepsilon' j_2' \rangle = \sqrt{\frac{2j_1''+1}{2j_1'+1}} \sqrt{\frac{2j_2''+1}{2j_2'+1}} \frac{1}{\sqrt{2(1+\delta_{k',0})2(1+\delta_{k'',0})}} \sum_{\lambda_1 \mu_1 \lambda_2 \lambda} \frac{v_{\lambda_1 \mu_1 \lambda_2 \lambda}(R)}{1+\delta_{\mu_1 0}} \sqrt{\frac{2\lambda_1+1}{4\pi}} \sqrt{\frac{2\lambda_2+1}{4\pi}} C_{j_2'',0,\lambda_2,0}^{j_2',0}$ $\times \left\{ \left(C_{j_1'',k'',\lambda_1,\mu_1}^{j_1',k'} + \varepsilon' C_{j_1'',k'',\lambda_1,\mu_1}^{j_1',-k'} + \varepsilon'' C_{j_1'',-k'',\lambda_1,\mu_1}^{j_1',k'} + \varepsilon' \varepsilon'' C_{j_1'',-k'',\lambda_1,\mu_1}^{j_1',-k'} \right) \right.$ $+ (-1)^{\lambda_1+\mu_1+\lambda_2+\lambda} \left(C_{j_1'',k'',\lambda_1,-\mu_1}^{j_1',k'} + \varepsilon' C_{j_1'',k'',\lambda_1,-\mu_1}^{j_1',-k'} + \varepsilon'' C_{j_1'',-k'',\lambda_1,-\mu_1}^{j_1',k'} + \varepsilon' \varepsilon'' C_{j_1'',-k'',\lambda_1,-\mu_1}^{j_1',-k'} \right) \left. \right\}$ $\times \left\{ \sum_{m_1=-j_1'}^{+j_1'} C_{j_1',m_1,j_2',m-m_1}^{j_1',m} \left(\sum_{\eta=-\min(\lambda_1,\lambda_2)}^{+\min(\lambda_1,\lambda_2)} C_{\lambda_1,\eta,\lambda_2,-\eta}^{\lambda,0} C_{j_1'',m_1-\eta,j_2'',m-(m_1-\eta)}^{j_1'',m} C_{j_1',m_1-\eta,\lambda_1,\eta}^{j_1',m_1} C_{j_2',m-(m_1-\eta),\lambda_2,-\eta}^{j_2',m-m_1} \right) \right\}$
8	asym. top + rigid diatom	$\langle j'' m j_1'' j_2'' V(\Lambda) j' m j_1' j_2' \rangle$ $= \sqrt{\frac{2j_1''+1}{2j_1'+1}} \sqrt{\frac{2j_2''+1}{2j_2'+1}} \sum_{\lambda_1 \mu_1 \lambda_2 \lambda} \frac{v_{\lambda_1 \mu_1 \lambda_2 \lambda}(R)}{1+\delta_{\mu_1 0}} \sqrt{\frac{2\lambda_1+1}{4\pi}} \sqrt{\frac{2\lambda_2+1}{4\pi}} C_{j_2'',0,\lambda_2,0}^{j_2',0}$ $\times \left\{ \sum_{k_1'=-j_1'}^{+j_1'} b_{k_1'} b_{k_1'-\mu_1} \left[C_{j_1'',k_1'-\mu_1,\lambda_1,\mu_1}^{j_1',k_1'} + (-1)^{\lambda_1+\mu_1+\lambda_2+\lambda} C_{j_1'',k_1'+\mu_1,\lambda_1,-\mu_1}^{j_1',k_1'} \right] \right\}$ $\times \left\{ \sum_{m_1=-j_1'}^{+j_1'} C_{j_1',m_1,j_2',m-m_1}^{j_1',m} \left(\sum_{\eta=-\min(\lambda_1,\lambda_2)}^{+\min(\lambda_1,\lambda_2)} C_{\lambda_1,\eta,\lambda_2,-\eta}^{\lambda,0} C_{j_1'',m_1-\eta,j_2'',m-(m_1-\eta)}^{j_1'',m} C_{j_1',m_1-\eta,\lambda_1,\eta}^{j_1',m_1} C_{j_2',m-(m_1-\eta),\lambda_2,-\eta}^{j_2',m-m_1} \right) \right\}$

9	sym. top + asym. top	$ \begin{aligned} \langle j'' m j_1'' j_2'' k_2'' \varepsilon'' V(\Lambda) j' m j_1' j_2' k_2' \varepsilon' \rangle &= \sqrt{\frac{2j_1''+1}{2j_1'+1}} \sqrt{\frac{2j_2''+1}{2j_2'+1}} \frac{1}{\sqrt{2(1+\delta_{k_2',0})2(1+\delta_{k_2'',0})}} \sum_{\lambda_1 \mu_1 \lambda_2 \mu_2 \lambda} \frac{v_{\lambda_1 \mu_1 \lambda_2 \mu_2 \lambda}(R)}{1+\delta_{\mu_1 0} \delta_{\mu_2 0}} \sqrt{\frac{2\lambda_1+1}{8\pi^2}} \sqrt{\frac{2\lambda_2+1}{8\pi^2}} \\ &\times \left\{ \sum_{m_1=-j_1'}^{+j_1'} C_{j_1', m_1, j_2', m-m_1}^{j', m} \sum_{\eta=-\min(\lambda_1, \lambda_2)}^{+\min(\lambda_1, \lambda_2)} C_{\lambda_1, \eta, \lambda_2, -\eta}^{\lambda, 0} C_{j_1'', m_1-\eta, j_2'', m-(m_1-\eta)}^{j'', m} C_{j_1'', m_1-\eta, \lambda_1, \eta}^{j_1', m_1} C_{j_2'', m-(m_1-\eta), \lambda_2, -\eta}^{j_2', m-m_1} \right\} \\ &\times \left\{ \sum_{k_1'=-j_1'}^{+j_1'} b_{k_1'} b_{k_1'-\mu_1} \left[\left(C_{j_1'', k_1', -\mu_1, \lambda_1, \mu_1}^{j_1', k_1'} \left(C_{j_2'', k_2', \lambda_2, \mu_2}^{j_2', k_2'} + \varepsilon' C_{j_2'', k_2', \lambda_2, \mu_2}^{j_2', -k_2'} + \varepsilon'' C_{j_2'', -k_2', \lambda_2, \mu_2}^{j_2', k_2'} + \varepsilon' \varepsilon'' C_{j_2'', -k_2', \lambda_2, \mu_2}^{j_2', -k_2'} \right) \right) \right. \right. \\ &\quad \left. \left. + (-1)^{\lambda_1+\mu_1+\lambda_2+\mu_2+\lambda} \left(C_{j_1'', k_1'+\mu_1, \lambda_1, -\mu_1}^{j_1', k_1'} \left(C_{j_2'', k_2', \lambda_2, -\mu_2}^{j_2', k_2'} + \varepsilon' C_{j_2'', k_2', \lambda_2, -\mu_2}^{j_2', -k_2'} + \varepsilon'' C_{j_2'', -k_2', \lambda_2, -\mu_2}^{j_2', k_2'} + \varepsilon' \varepsilon'' C_{j_2'', -k_2', \lambda_2, -\mu_2}^{j_2', -k_2'} \right) \right) \right] \right\} \end{aligned} $
0	asym. top + asym. top	$ \begin{aligned} \langle j'' m j_1'' j_2'' V(\Lambda) j' m j_1' j_2' \rangle &= \sqrt{\frac{2j_1''+1}{2j_1'+1}} \sqrt{\frac{2j_2''+1}{2j_2'+1}} \sum_{\lambda_1 \mu_1 \lambda_2 \mu_2 \lambda} \frac{v_{\lambda_1 \mu_1 \lambda_2 \mu_2 \lambda}(R)}{1+\delta_{\mu_1 0} \delta_{\mu_2 0}} \sqrt{\frac{2\lambda_1+1}{8\pi^2}} \sqrt{\frac{2\lambda_2+1}{8\pi^2}} \\ &\left\{ \left(\sum_{k_1'=-j_1'}^{+j_1'} b_{k_1'} b_{k_1'-\mu_1} C_{j_1'', k_1', -\mu_1, \lambda_1, \mu_1}^{j_1', k_1'} \right) \left(\sum_{k_2'=-j_2'}^{+j_2'} b_{k_2'} b_{k_2'-\mu_2} C_{j_2'', k_2', -\mu_2, \lambda_2, \mu_2}^{j_2', k_2'} \right) \right. \\ &\quad \left. + (-1)^{\lambda_1+\mu_1+\lambda_2+\mu_2+\lambda} \left(\sum_{k_1'=-j_1'}^{+j_1'} b_{k_1'} b_{k_1'+\mu_1} C_{j_1'', k_1'+\mu_1, \lambda_1, -\mu_1}^{j_1', k_1'} \right) \left(\sum_{k_2'=-j_2'}^{+j_2'} b_{k_2'} b_{k_2'+\mu_2} C_{j_2'', k_2'+\mu_2, \lambda_2, -\mu_2}^{j_2', k_2'} \right) \right\} \\ &\times \sum_{m_1=-j_1'}^{+j_1'} C_{j_1', m_1, j_2', m-m_1}^{j', m} \left(\sum_{\eta=-\min(\lambda_1, \lambda_2)}^{+\min(\lambda_1, \lambda_2)} C_{\lambda_1, \eta, \lambda_2, -\eta}^{\lambda, 0} C_{j_1'', m_1-\eta, j_2'', m-(m_1-\eta)}^{j'', m} C_{j_1'', m_1-\eta, \lambda_1, \eta}^{j_1', m_1} C_{j_2'', m-(m_1-\eta), \lambda_2, -\eta}^{j_2', m-m_1} \right) \\ \text{If IDENTICAL_PES=YES, combine with:} \\ &+ \left\{ \left(\sum_{k_1'=-j_1'}^{+j_1'} b_{k_1'} b_{k_1'-\mu_2} C_{j_1'', k_1', -\mu_2, \lambda_2, \mu_2}^{j_1', k_1'} \right) \left(\sum_{k_2'=-j_2'}^{+j_2'} b_{k_2'} b_{k_2'-\mu_2} C_{j_2'', k_2', -\mu_1, \lambda_1, \mu_1}^{j_2', k_2'} \right) \right. \\ &\quad \left. + (-1)^{\lambda_1+\mu_1+\lambda_2+\mu_2+\lambda} \left(\sum_{k_1'=-j_1'}^{+j_1'} b_{k_1'} b_{k_1'+\mu_2} C_{j_1'', k_1'+\mu_2, \lambda_2, -\mu_2}^{j_1', k_1'} \right) \left(\sum_{k_2'=-j_2'}^{+j_2'} b_{k_2'} b_{k_2'+\mu_1} C_{j_2'', k_2'+\mu_1, \lambda_1, -\mu_1}^{j_2', k_2'} \right) \right\} \\ &\times \sum_{m_1=-j_1'}^{+j_1'} C_{j_1', m_1, j_2', m-m_1}^{j', m} \left(\sum_{\eta=-\min(\lambda_1, \lambda_2)}^{+\min(\lambda_1, \lambda_2)} C_{\lambda_2, \eta, \lambda_1, -\eta}^{\lambda, 0} C_{j_1'', m_1-\eta, j_2'', m-(m_1-\eta)}^{j'', m} C_{j_1'', m_1-\eta, \lambda_2, \eta}^{j_1', m_1} C_{j_2'', m-(m_1-\eta), \lambda_1, -\eta}^{j_2', m-m_1} \right) \end{aligned} $

Notes: In the code, for efficiency, the calculation of Clebsch-Gordon coefficients is precluded by a check of triangle rule, to avoid computing those coefficients that are zero by definition.

For asymmetric tops the range of summation over k'_1 is either $-j'_1 + \mu_1 \leq k'_1 \leq j'_1$ or $-j'_1 \leq k'_1 \leq j'_1 - \mu_1$, depending on the sign of μ_1 (in the formula above this is not indicated explicitly, for transparency, but is implicit in the code). And the same for k'_2 .

Basis set expansion for `SYS_TYPE=2, 6` (that include vibrational motion) is not implemented in this release of the code but will probably be implemented in the future. Meanwhile, users can use Option 1 to compute matrix elements (by direct integration).

VIII. Sampling of Initial Conditions and Monte-Carlo Simulations

By default, the initial conditions for MQCT trajectories are set-up for all integer values of the orbital angular momentum quantum number ℓ in the range $0 \leq \ell \leq \ell_{\max}$, with $\ell_{\max}$ determined either by the maximum value of total angular momentum and the angular momentum of the initial state as $\ell_{\max} = J_{\max} - j$, or by the maximum impact parameter and collision energy as $\ell_{\max} = b_{\max} \sqrt{2\mu U} / \hbar$. The results of calculations are summarized in three files as follows: transition probabilities (summed over the states of each final channel) from the individual states of the initial channel to all channels are printed in the file `OPACITY_FUNCTIONS_L.out`, ordered according to the channels list, whereas the total elastic and inelastic probabilities are printed in the file `TOTAL_PROBABILITIES_L.out`, as dependencies on the impact parameter b and the orbital angular momentum l . Partial cross-sections for each channel (transition probabilities multiplied by $2J + 1$) are printed in the file `PARTIAL_CROSS_SECTIONS_J.out` as a function of total angular momentum J . First few lines of each file contain information about the initial state (state number in the list, j and m), the collision energy U and the header line, followed by the table of results. Results are printed with the Billing energy correction applied to partial cross-sections. The values of deflection angle of MQCT trajectories and the phase shift obtained from them are printed in the file `DEFLECTION_FUNCTION.out`. These files are generated by default. We recommend that users inspect all these dependencies after each program's execution, to make sure that the overall behavior of the system is reasonable.

These dependencies are often smooth, so, the users have an option of skipping some values of ℓ using a keyword `DL` to make the overall calculations more affordable. For example, if `DL=3` is indicated, the code will only do calculations for $\ell = 1, 4, 7, 10, \dots$ (one value of ℓ in the middle of each interval) and then will interpolate all results to fill the gaps. In those cases when the behavior of opacity function is different in the short-range, mid-range and long-range interaction regions, users may want to use up to three different values of `DL`. For example, if one wants to use `DL=3` in the range $b \leq 10$ Bohr and then switch to `DL=20` in the range $10 \leq b \leq 30$ Bohr, user should indicate `DL=3, DL_LR=20, B_LR=10, B_MAX=30`. For more accurate description of the mid-range one can indicate `DL=3, DL_MR=1, B_MR=6, DL_LR=20, B_LR=10, B_MAX=30`. It is better to use odd number for `DL`, so that the ℓ value in the center of each `DL` interval is sampled and propagated. For the case of `IDENTICAL=YES` (see below) the values of

DL must be odd, since only in this case both even and odd values of ℓ will be sampled and propagated (important for identical collision partners). The default “interpolation” method is simply a rectangular quadrature, but the optional keyword `PROB_SPLINE` can be used for cubic spline interpolation. In this case the code splines the logarithm of opacity functions and then exponentiates the spline. Deflection functions are processed as they are. The result of spline for one initial (j, m) -state and all final channels is saved to the file `SPLINED_DEFLECTION_OPACITY_L.out` for the user to check the quality and behavior of spline. The values of j , m and, when appropriate, the parity for this state can be indicated using keywords `PRN_JM` and `PRN_P`.

For heavier molecules and higher collision energies (large $J_{\max}$ or $\ell_{\max}$) and/or for highly excited rotational states (large j_1 and j_2), the number of initial states covered may become prohibitively expensive (large j , many values of m). Running MQCT trajectories for all possible initial degenerate states, which is a default in our code, may become computationally expensive and, in fact, is unnecessary. In such cases, it is more efficient to sample the values of ℓ , j and m randomly and simultaneously, using an efficient multi-dimensional Monte-Carlo procedure. This option is initiated by the following keywords: `MONTE_CARLO=YES`, `NMB_TRAJ=200`, where `NMB_TRAJ` is the number of trajectories to compute. Note that in the case of Monte-Carlo sampling the elastic cross section is computed only approximately (it depends on the choice of $b_{\max}$ and should be ignored), while the differential cross section is not computed at all (due to the lack of accurate phase information). The random sampling procedure implemented in this version of MQCT is somewhat different from the previously published [Mandal 2021]. For all molecule + molecule system types it starts from sampling the initial value of j through the range $|j_1 - j_2| \leq j \leq j_1 + j_2$. This sampling is weighted, to give proportionally more weight to higher values of j that have more m states (by making these boxes larger). For the molecule + atom system types, this step is not needed since the value of j is uniquely determined by the initial channel. Second step is sampling of the initial value of m through the range $0 \leq m \leq j$. This sampling is again weighted, to give twice more weight to $m \geq 1$ states (relative to $m = 0$) in order to account for negative values of m (by making these boxes twice larger). Finally, the value of ℓ is sampled uniformly through the range $0 \leq \ell \leq \ell_{\max}$. This procedure replaces a multi-dimensional sampling with one Monte-Carlo sampling, which is very efficient. All properties associated with individual trajectories in the Monte-Carlo sample are provided in the file `MC_InitialCond.DAT` which includes trajectory number, initial state index, corresponding values of j and m , parity of the initial state (relevant to the case of identical collision partners), the value of ℓ and the weight of each trajectory. Here, the weight is the frequency of a particular trajectory being sampled (number of hits, can be 0 or 1 for $m = 0$ states and 0, 1 or 2 for $m \geq 1$ states, which accounts for positive and negative values of m). In the cases when `NMB_TRAJ` indicated by user is greater or equal to the total number of trajectories possible for given initial channel and $\ell_{\max}$, the weights are set to their asymptotic values ($w = 1$ for $m = 0$ and $w = 2$ for $m \geq 1$) and total number of hits is adjusted to the total number of trajectories possible. By referring to this file users can obtain all details about the sampling process. The results of Monte-Carlo calculations are summarized in three output files as follows: Inelastic transition probabilities for individual trajectories are printed in the file `MC_Probabilities.out`. Evolution of Monte-Carlo cross-sections for each inelastic channel, as more trajectories are added to the sample, is printed in the file `MC_Convergence.out`. The `MC_Statistics.out` file contains information regarding the evolution of statistical error estimates for all inelastic cross sections, as more trajectories are added to the sample. The maximum

statistical error observed among all inelastic channels is written into the main output file `CROSS_SECTIONS.out`, together with the value of Monte Carlo coverage, which represents the percentage of sampled trajectories relative to the total number of trajectories possible for given initial channel and $\ell_{\max}$. If additional trajectories are needed (to reduce statistical error), users can raise the value of `NMB_TRAJ` and run the calculations one more time. The code automatically adds new trajectories to the sample and updates all the output files. Often 100 trajectories are sufficient for a semi-quantitative prediction of cross sections. Monte-Carlo calculations with 200 trajectories give results within ten percent of the regular MQCT calculations.

Calculations of differential cross sections involves phase information and computation of a coherent sum over all partial scattering waves, and thus requires propagation of MQCT trajectories for *all* allowed integer values of ℓ , which is default in the code [Mandal 2018]. The keyword `DIFF_CROSS=YES` can be used to request construction of the differential cross section. Angular resolution of the differential cross section is defined by `ANG_RES`. Angular dependence of differential cross section is printed into the file `DIFF_CROSS.out`.

However, when the Monte-Carlo sampling of the initial conditions is requested (for numerical efficiency, see above) the values of ℓ will be chosen randomly and only a few of them may be available for each initial degenerate state jm . In this case the meaningful determination of the deflection function, scattering phase and the differential cross section is technically challenging, and is not implemented in the code. So, `MONTE_CARLO=YES` should not be used simultaneously with calculations of the differential cross section. Same considerations apply to the integral cross section for the elastic scattering channel, since it also requires the scattering phase. If the `MONTE_CARLO=YES` option is used, the value of elastic cross section should normally be ignored as explained above (zero is printed in the output file). In many applications, the elastic and/or differential cross section are not needed. Then `MONTE_CARLO=YES` is the best option.

If the elastic and/or differential cross sections are needed for complex systems, the most efficient and robust approach is to use the optional keyword `DL`, in order to skip some values of ℓ and thus make the overall calculations more affordable. For calculation of the differential cross section the code still computes a coherent sum over all ℓ , but within the “boxes” of size `DL` the same values of scattering phase and probability amplitudes are used. The magnitude of `DL` becomes a convergence parameter in this case and should be carefully checked by convergence studies.

There are several options in the code that allow users to visualize MQCT trajectories, or their most important properties. Opacity functions and deflection functions are printed by default, which gives dependencies of transition probabilities and scattering angle on ℓ and b (but only for `MONTE_CARLO=NO`, which is default). Another useful option is to use keyword `PRN_TRJCT` to print out all information for a trajectory with indicated value of ℓ . By default, the information from trajectories with the initial $j = \max(j_1, j_2)$ and $m = 0$ is printed. Other values of the initial j and m can be specified using optional keyword, for example: `PRN_JM=3, 3`. Plotting and inspecting the deflection function, opacity functions and some trajectories is recommended, particularly at low scattering energies, when trajectories may be trapped at certain values of ℓ .

To monitor the strength of collision, we introduced the `PRN_MTRX` keyword, which prints the evolution of some matrix elements along the trajectory, as a function of time. Specific elements to be printed should be listed using `TRANS_LST` keyword. Arguments for this list correspond to the `k_matel` indexes of matrix elements, the unique transition identifiers stored in the `MTRX.DAT` (serial) or `MTRX_INFO.DAT` (parallel) files.

There are options in the code to deal automatically with trapped trajectories, if those occur. For such trajectories the number of loops (due to mutual rotation of collision partners around the origin), and the number of periods (due to mutual oscillations of collision partners along the intermolecular distance R) is determined. The option `NMB_LOOPS=2` commands to stop propagation of orbiting trajectories after two full loops (the default value is 1). The option `NMB_OSCIL=5` commands to stop propagation at the fifth outer turning point (the default value is 1). When such trapped trajectory is forced to stop, it is still analyzed in a standard way. However, the result of such analysis is somewhat arbitrary, since the termination time is also arbitrary. The option `NO_RESONANCE=YES` tells the code to remove looping and oscillating trajectories from analysis, which can be used for calculation of non-resonant contribution to the integral cross sections.

Note that differential cross sections and the elastic channel integral cross sections can't be computed rigorously at collision energies when at least one resonant trajectory is present, since in such cases the deflection function is undefined in the range of small values of ℓ . A good recipe for extracting resonance information from trapped trajectories is yet to be found, but AT-MQCT helps in this respect. In any case the code automatically detects trajectories that exhibit resonant behavior and prints some basic information about these trajectories in the file `RESONANCE_TRAJECT.out`, including the relevant values of ℓ . Then, if desired, the user can rerun the code with the option `PRN_TRJCT` employed, to obtain more detailed information for each resonant trajectory.

IX. Convergence Studies and Propagation of MQCT Equations

All output files have extension `*.out`. System setup is written into the file `USER_INPUT_CHECK.out` and should be checked by user for correctness. The file `STATES.out` (written if the option `PRINT_STATES=YES` is chosen) contains the list of all quantum states involved in calculations, including their quantum numbers and a state number in the overall list, assigned by MQCT. Cross sections for transitions from the initial state to the final states (all states of the basis, including the elastic channel) are listed in the output file `CROSS_SECTIONS.out`, for each effective collision energy specified in the input. For each transition, the actual collision energy `E_COLL` is also given in the output file, which depends on the effective collision energy `U_ENERGY` and the state-to-state energy difference. These are different for different transitions, particularly at lower collision energies, which is a property of the mixed quantum/classical approach, discussed in detail in several recent papers (the so-called Billing correction). Next in the output file goes an important information about the largest values of energy conservation error, and the probability conservation error (both given as % of the initial value) encountered during the propagation. Users should check these numbers to ensure that they are reasonable (say below

1.0E-03). Excessive values indicate that modification of the propagation parameters in the input file is needed. The last in the output file is CPU-time statistics of the code execution. More detailed information about propagation accuracy can be found in the file `INTEGRATOR_ERRORS.out`, where the data collected for the less accurate (worst) trajectory in the batch are printed, for each collision energy. Note that the energy conservation error is computed and printed by default but can be disabled using a keyword `ENERGY_ERROR=NO` in order to reduce numerical effort, particularly in CS-MQCT and AT-MQCT calculations where the total energy is not conserved anyway.

Convergence studies with respect to several input parameters should be carried out by the user. For the quantum part of the system, sensitivity of results (cross section values) should be checked with respect to the basis set size `NMB_CHNLS`, the range of molecule-molecule distances used to compute transition matrix elements `RMIN` and `RMAX`, the corresponding number of points `GRD_R`, and the number of integration points for each internal degree of freedom (parameters `GRD_ANG1`, `GRD_ANG2` and `GRD_ANG3`). If the PES is represented by expansion over analytic functions, then the convergence parameter is the number of terms `NMB_TERMS` (and, of course, what terms are included). For trajectory propagation, sensitivity of energy and probability conservation errors should be checked with respect to step size `TIME_STEP`, and, if the adaptive step-size is used, the value of tolerance `EPS_ODEINT` and the minimum time step `MIN_TMSTP`. For sampling of the initial conditions, convergence parameter is the upper limit of total angular momentum `JTOTU` (or, optionally, the maximum value of collision impact parameter `B_IMPCT`). If optional Monte-Carlo sampling of the initial states is chosen (`MONTE_CARLO=YES`) then the number of trajectories `NMB_TRAJ` is also a convergence parameter. The value of `RMAX` may also affect MQCT trajectories, since it is used to set up the initial molecule-molecule separation. Note that during trajectory propagation the value of `RMAX` can be specified as larger than the range covered by the grid of precomputed matrix elements. In this case, the code will automatically set to zero all matrix elements in the asymptotic range. If trajectory penetrates the range of distances smaller than `RMIN` (e.g., due to high collision energy) a linear function is used for extrapolation of matrix element into the repulsive short-distance range.

Note that during trajectory propagation the value of `RMAX` can be specified as larger than the range covered by the grid of precomputed matrix elements. In this case, the code will automatically extrapolate all matrix elements in the asymptotic range using a C_6/R^6 decay function with the value of C_6 using the value at the last point of the grid. If trajectory penetrates the range of distances smaller than `RMIN` (e.g., due to high collision energy), a linear extrapolation function is used based on the derivative at the first two grid points to extend matrix elements into the repulsive short-distance range.

Often, user will want to increase the number of channels relative to the previously run calculations, for example, in order to check convergence with respect to the basis set size, or to do calculations at higher collision energy, where the number of channels is typically larger. In this case our code allows us to add new elements needed to the existing transition matrix, without re-computing the entire matrix. This is achieved, simply, by rerunning the code with new increased number of channels, using any method of channel

specification available, for example, increasing the value of keyword `NMB_CHNLS` and adding new channels to the list `CHNLS_LIST` (or increasing the value of keyword `EMAX`). Users need to indicate `READ_MTRX=YES` so that the code will read the existing matrix from file, check whether all needed matrix elements are present, automatically compute the missing elements, update the matrix file and/or proceed with trajectory calculations, depending on what options are indicated by the user. One requirement is that *R*-grid remains identical to the one used in the previous calculations. However, the way of computing elements of the state-to-state transition matrix can be changed, for example, by increasing the number of integration points for the internal coordinates, or increasing the number of terms in the PES expansion, etc. The only requirement is that new channels are added at the end of the list, not at the beginning or stuffed in the middle (automatically taken care of if `EMAX` is used).

Also, the user can always run calculations of dynamics with the number of channels smaller than that in the saved matrix file `MTRX_UF.dat`. The code will read the file and choose only those matrix elements that are necessary for this run. Again, the requirement is that the active channels are listed continuously from the beginning of the list, and only the channels at the end of the list can be omitted. If the user wants to exclude some states from the beginning or from the middle of the list, the keyword `EXCLUDE_STATES` can be used as explained in Sec. VI (the corresponding data file should be provided).

Besides the default RK4 propagator, one can choose an adaptive step-size method by indicating:

```
PROPAGATOR=ODEINT, TIME_STEP=500.0, MIN_TMSTP=10.0, EPS_ODEINT=1.0E-3
```

This method adjusts time-steps along each trajectory trying to keep accuracy below `EPS_ODEINT` and is a slightly modified version of the code from Numerical Recipes. Our version of this propagator enforces `MIN_TMSTP` to avoid an excessively long integration near the turning point. We recommend that users run their first calculations using the default RK4 propagator to determine a suitable value of `TIME_STEP` for their system and collision conditions (by monitoring the conservation of energy and norm in the output file). Then, one can try to switch to the `ODEINT` by simultaneously setting `MIN_TMSTP` equal to this found value and increasing the value of `TIME_STEP` by an order of magnitude or even more (which will play the role of the maximum time step allowed in `ODEINT`). We found that for the systems with deep molecule-molecule attraction potentials, such as dipole-dipole, this approach gives a considerable computational advantage. Also, our `ODEINT` propagator has an option of `SINGLE_STEP` that commands to integrate the entire trajectory in one step. In this case the code automatically estimates a termination time for each trajectory (based on the impact parameter, the initial separation of collision partners, and the collision energy) and uses this number as maximum time step. This is the most efficient propagation option recommended for the production runs. Note that if `SINGLE_STEP` is chosen, then no propagation information is printed along the trajectory, no phases are computed, and thus no elastic or differential cross sections are calculated.

X. Molecular Symmetry and Identical Collision Partners

Rotational states of asymmetric-top molecules are split into two groups, called *para*- and *ortho*-states. We define them based on what values of the quantum number k participate in the expansion of wave function over the basis of symmetric-top eigenstates (expansion coefficients b_k). Namely, for each j , *even* values of k produce *para*-states, while *odd* values of k produce *ortho*-states. Even and odd values of k never mix in the same $j_{k_a k_c}$ state. For symmetric molecules, such as H₂O, transitions are allowed within each group only, and are exactly forbidden between the *para*- and *ortho*-states, due to the symmetry of potential of interaction of the molecule with a quencher (any quencher). Including all states would not cause a problem but would be meaningless since the efficiency of calculations would be reduced. (However, including *ortho*- and *para*-states in the same calculations can be done as a test of numerical convergence: if calculations are accurate, these forbidden transitions should have numerically negligible matrix elements and cross sections; if they are not, that something is wrong either with PES interface, or with numerical convergence of calculations.)

If the states are specified explicitly as a list, the user should take care of this issue manually. But, if the basis set is generated automatically (e.g., using the keyword EMAX), user has an option to indicate SYMMETRY=YES, for reducing the basis set size to one symmetry only, depending on symmetry of the initial state (as indicated in the input file). In the molecule + molecule case, symmetry consideration will be applied to the states of each molecule individually, for example, in a system like H₂O + H₂ one can run *four* sets of independent calculations for *p*-H₂O + *p*-H₂, *p*-H₂O + *o*-H₂, *o*-H₂O + *p*-H₂ and *o*-H₂O + *o*-H₂. Note, however, that in asymmetric molecules, such as HDO, all transitions are allowed, and all states of HDO should be included in the basis. Thus, indicating the keyword SYMMETRY=YES, would lead to unphysical results in this and other cases with no symmetry, and should not be done.

Ortho and *para* states of asymmetric top rotor are identified by the code (by analyzing the expansion coefficients b_k) and are given assignments: $\kappa = 0$ for *para* states and $\kappa = 1$ for *ortho* states (here we use Greek symbol kappa). They can easily be distinguished by users too, based on $j_{k_a k_c}$ assignment of a state: for *para* states the sum $k_a + k_c$ is even (e.g., 0₀₀, 1₁₁, 2₂₀, etc.) while for *ortho* states the sum $k_a + k_c$ is odd (e.g., 1₁₀, 1₀₁, 2₁₀, etc.).

Another symmetry property of a molecular state is its *inversion parity*, determined by the signs of two expansion coefficients b_k that correspond to two symmetric-top basis functions with the same absolute value of $|\pm k|$. For *positive* parity states, the two components have the *same* expansion coefficients, $b_{-k} = b_{+k}$ (which corresponds to constructive superposition, “in phase”), while for *negative* parity they have *opposite* signs: $b_{-k} = -b_{+k}$ (destructive superposition, “out of phase”). This is equivalent to the inversion parity of symmetric-top functions defined in Sec. I above, so, we can write $b_{-k} = \varepsilon b_{+k}$ where $\varepsilon = \pm$ is parity. Our code determines the inversion parity of each state (by analyzing the signs of $b_{\pm k}$). In the file USER_INPUT_CHECK.out user can find the assignments of asymmetric-top rotor states in terms of inversion parity and *ortho/para* character.

When two identical molecules are collided (e.g., two diatomics in SYS_TYPE=5, 6, e.g., CO + CO or two asymmetric-top rotors in SYS_TYPE=0, e.g., H₂O + H₂O) one may want to treat them as *indistinguishable*, by indicating the keyword

IDENTICAL=YES in the input file. In this case only one set of rotational constants, and only the unique channels should be specified. For example, if the state $(j_{1k_{a_1}k_{c_1}}, j_{2k_{a_2}k_{c_2}}) = (1_{11}, 0_{00})$ of $\text{H}_2\text{O} + \text{H}_2\text{O}$ is already specified, one should not include the state $(0_{00}, 1_{11})$ since two molecules are treated as indistinguishable. However, for a system of two indistinguishable molecules we define two symmetrized wavefunctions of the system, those with positive and negative *exchange parities* ($\pm$):

$$|jmj_1j_2\rangle^\pm = \frac{|jmj_1j_2\rangle_{\Lambda_1, \Lambda_2} \pm |jmj_1j_2\rangle_{\tilde{\Lambda}_2, \tilde{\Lambda}_1}}{\sqrt{2(1 + \delta_{j_1j_2})}}$$

The second term in this expression (subscript $\tilde{\Lambda}_2, \tilde{\Lambda}_1$) is obtained from the first one (subscript Λ_1, Λ_2) by swapping two collision partners (molecules 1 and 2). Such symmetrized states are energetically degenerate but transitions between them are forbidden. If the keyword IDENTICAL=YES is indicated in the input file, MQCT code will carry out two consecutive independent calculations of collision dynamics, one with the states of positive *exchange parity* $|jmj_1j_2\rangle^+$, and second with the states of negative *exchange parity* $|jmj_1j_2\rangle^-$ and will average the results of two calculations using the weighting factor indicated by keyword WGH_T_POSPAR (as explained below).

It is possible to show that this expression can be rewritten in the following form:

$$|jmj_1j_2\rangle^\pm = \frac{|jmj_1j_2\rangle_{\Lambda_1, \Lambda_2} \pm p |jmj_2j_1\rangle_{\Lambda_1, \Lambda_2}}{\sqrt{2(1 + \delta_{j_1j_2})}}$$

where the *overall inversion parity* of the rotational state of *two* asymmetric-top rotors is introduced as $p = (-1)^j(-1)^{\kappa_1+\kappa_2}\varepsilon_1\varepsilon_2$, where κ_1 and κ_2 correspond to the para/ortho character (0 or 1) of the two states, while ε_1 and ε_2 correspond to their inversion parities (± 1). Note that j_1 and j_2 are swapped in the second term, which is a shortened notation that means that we use quantum state 2 for molecule 1, whereas quantum state 1 is used for molecule 2 (this includes all quantum numbers $j_{k_a k_c}$ and the corresponding set of expansion coefficients b_k). Similar, $\delta_{j_1j_2} = 1$ only if two molecules are in the same quantum state $j_{k_a k_c}$ with the same set of expansion coefficients b_k . For a system of two linear-top rotors the expression of overall inversion parity is simpler: $p = (-1)^j$.

Note that since the definition of the overall inversion parity p includes $(-1)^j$, and since the value of j is varied through the range $|j_1 - j_2| \leq j \leq j_1 + j_2$, different j -states within each channel (j_1, j_2) indicated in the input file, will correspond to different inversion parities p . Therefore, in general, the states with both positive and negative values of p will be found within any initial channel (except the trivial case when, say, $j_1 = 0$, and therefore only one value of $j = j_2$ is possible).

Special cases occur if the states of two particles are the same, such as $(0_{00}, 0_{00})$ or $(1_{11}, 1_{11})$. Then, only one exchange parity state is possible for each j (and, in fact, this state is equivalent to the original state of distinguishable collision partners), because the other parity annihilates the overall wavefunction. Since for these states $\kappa_1 = \kappa_2$ and $\varepsilon_1 = \varepsilon_2$, the expression for parity simplifies into $p = (-1)^j$. Consider $(1_{11}, 1_{11})$ channel as example. It has three components, $j = 0, 1$ and 2 . For even states $j = 0$ and 2 we have $p = +1$ for inversion parity and therefore only positive exchange parity is meaningful. In contrast, for $j = 1$ we have $p = -1$ for the

inversion parity and therefore only negative exchange parity is meaningful. MQCT code recognizes these cases and takes care of them automatically, so that only non-vanishing states are included in the rotational basis (and only one set of calculations with the meaningful exchange parity is carried out for each j state within the initial channel, see below). One may say that within these channels the states with various possible j values are split onto two groups: one group shows up in the positive, whereas the other shows up in the negative exchange parity blocks of the transition matrix (see below). One exception is the trivial case of $(0_{00}, 0_{00})$ where only one value of the overall j is possible ($j = 0$) which belongs to the positive exchange parity block. In general cases, such as $(0_{00}, 1_{11})$, each j state will be present in both positive and negative exchange parity blocks of the transition matrix.

If transition matrix is calculated by direct integration (Option 1), the code will symmetrize wavefunctions as explained above and will compute matrix elements for state-to-state transitions within each exchange parity block. But, if transition matrix is calculated by PES expansion (Option 2), the code will build matrix elements by combining terms that correspond to distinguishable collision partners (without symmetrization of wavefunctions), which produces a symmetrized matrix for calculations within positive and negative exchange parities, as follows:

$$\langle j'' m_{j_1''} j_2'' | V(\Lambda_1, \Lambda_2) | j' m_{j_1'} j_2' \rangle^\pm = \frac{1}{\sqrt{2(1 + \delta_{j_1', j_2'}) 2(1 + \delta_{j_1'', j_2''})}} \begin{bmatrix} \langle j'' m_{j_1''} j_2'' | V(\Lambda_1, \Lambda_2) | j' m_{j_1'} j_2' \rangle \\ \pm p' \times \langle j'' m_{j_1''} j_2'' | V(\Lambda_1, \Lambda_2) | j' m_{j_2'} j_1' \rangle \\ \pm p'' \times \langle j'' m_{j_2''} j_1'' | V(\Lambda_1, \Lambda_2) | j' m_{j_1'} j_2' \rangle \\ + p' p'' \times \langle j'' m_{j_2''} j_1'' | V(\Lambda_1, \Lambda_2) | j' m_{j_2'} j_1' \rangle \end{bmatrix}$$

Here p' and p'' represent the overall inversion parity (defined above) for the initial and final states of a two-molecule system. The symmetry of interaction potential permits to simplify it, as follows (using permutations of the initial state):

$$\langle j'' m_{j_1''} j_2'' | V(\Lambda_1, \Lambda_2) | j' m_{j_1'} j_2' \rangle^\pm = \frac{1}{\sqrt{(1 + \delta_{j_1', j_2'})(1 + \delta_{j_1'', j_2''})}} \begin{bmatrix} \langle j'' m_{j_1''} j_2'' | V(\Lambda_1, \Lambda_2) | j' m_{j_1'} j_2' \rangle \\ \pm p' \langle j'' m_{j_1''} j_2'' | V(\Lambda_1, \Lambda_2) | j' m_{j_2'} j_1' \rangle \end{bmatrix}$$

This is implemented in the code. Analysis of this expression shows that in the case of indistinguishable collision partners transitions are possible only between the states of the same exchange parity ($\pm$). Therefore, if IDENTICAL=YES is chosen, MQCT code will run two independent calculations: one with $\langle j'' j_1'' j_2'' | V | j' j_1' j_2' \rangle^+$ and another with $\langle j'' j_1'' j_2'' | V | j' j_1' j_2' \rangle^-$ transition matrixes. If the initial channel is characterized by $j_1' \neq j_2'$, the code will run two sets of calculations for each state j within the initial channel, but in the special case of $j_1' = j_2'$ the code will run only one set of calculations for each j' of the initial channel. Namely, calculations for even initial j' will use $\langle j'' j_1'' j_2'' | V | j' j_1' j_2' \rangle^+$ whereas calculations for odd initial j' will use $\langle j'' j_1'' j_2'' | V | j' j_1' j_2' \rangle^-$, see above.

It should also be noted that since the translational motion of collision partners is treated classically in MQCT, the state-to-state transition matrix does not depend on the angular momentum quantum number ℓ . However, in the case of identical collision partners

(IDENTICAL=YES) one can introduce the effect of even and odd ℓ values, to mimic the full-quantum approach where the factor $(-1)^\ell$ is included. This is done as follows: In a general case of $j'_1 \neq j'_2$ the calculations for each j' are done with all values of ℓ , both even and odd. Trajectories from the first run with matrix $\langle j''j'_1j'_2|V|j'j'_1j'_2\rangle^+$ are split to compute $\sigma_{\text{evn}}^{(+)}$ and $\sigma_{\text{odd}}^{(-)}$, depending on their values of ℓ , whereas trajectories from the second run with matrix $\langle j''j'_1j'_2|V|j'j'_1j'_2\rangle^-$ are split to compute $\sigma_{\text{evn}}^{(-)}$ and $\sigma_{\text{odd}}^{(+)}$. Those are combined to compute the weighted average $\sigma = w\sigma^{(+)} + (1-w)\sigma^{(-)}$, where $\sigma^{(+)} = \sigma_{\text{evn}}^{(+)} + \sigma_{\text{odd}}^{(+)}$ and $\sigma^{(-)} = \sigma_{\text{evn}}^{(-)} + \sigma_{\text{odd}}^{(-)}$, while w is the value of keyword WGHT_POSPAR that user should provide in the input file to account for the nuclear spin weights of two exchange parities. The default value is WGHT_POSPAR=1.d0, which leads to the overall cross section $\sigma = \sigma^{(+)}$. In this case the code will only need trajectories with even values of ℓ from the first run and with odd values of ℓ from the second run. The value of WGHT_POSPAR=0.d0, in contrast, will result in $\sigma = \sigma^{(-)}$, in which case the code will only need trajectories with odd values of ℓ from the first run and with even values of ℓ from the second run (for each initial j').

The special case of $j'_1 = j'_2$, when only one run of calculations is done for each j' , is treated similar, namely: For even initial j' one run of calculations is done with matrix $\langle j''j'_1j'_2|V|j'j'_1j'_2\rangle^+$ which gives $\sigma_{\text{evn}}^{(+)}$ and $\sigma_{\text{odd}}^{(-)}$, while for odd initial j' one run of calculations is done with matrix $\langle j''j'_1j'_2|V|j'j'_1j'_2\rangle^-$ which gives $\sigma_{\text{evn}}^{(-)}$ and $\sigma_{\text{odd}}^{(+)}$. Note that in these runs all values of ℓ are sampled, but then all trajectories are split into those with even and odd ℓ , just as in the general case. However, since now only one run is done per j' , the number of trajectories that contribute to each cross section is smaller (either even ℓ or odd ℓ), which makes these cross sections smaller by a factor of ~ 2 . The weighted average is computed as in the general case and the reduction of sampled trajectories to only even or only odd ℓ values is possible only in the special cases of $w = 1$ or $w = 0$ as explained above.

Therefore, if in the case of identical collision partners users want to skip some values of ℓ using the keyword DL, they should use odd values of DL, since only in this case both even and odd values of ℓ will be sampled and propagated. If the Monte-Carlo sampling of ℓ is employed (see above), the exchange parity ($\pm$) is also sampled randomly as a part of the multi-dimensional sampling procedure.

Ortho- and para-symmetry considerations remain valid within each exchange parity block of states ($\pm$) within the transition matrix. If the basis set is specified explicitly as a list of states, the user should make efforts to exclude the uncoupled states. If the basis set is generated automatically (employing the keyword EMAX, or keywords EMAX1 and EMAX2), user has option to minimize the number of channels by choosing SYMMETRY=YES. Again, only the states coupled to the initial state would be included, based on ortho- and para-character of the states. It also follows that for the system of two indistinguishable molecules in the cases of para + para and ortho + ortho the elements of symmetrized transition matrix are different for two exchange parities ($\pm$) and are different from those of distinguishable particles case (thus, the keyword IDENTICAL=YES is required for the code to do two consecutive runs and compute the average). However, in the cases of para + ortho and ortho + para the symmetrized matrix is the same for two exchange parities and

is exactly equivalent to the case of non-identical collision partners. So, for these subsets of states the calculations can and should be done without the keyword IDENTICAL=YES.

Finally, it should be noted that the last version of MQCT has the factor $(1 + \delta_{j_1 j_2})(1 + \delta_{j'_1 j'_2})$ included automatically in the calculations of cross sections for system types 5, 6 and 0 consistent with the paper of Green [JCP **62**, 2271 (1975)], appropriate for conversion into rate coefficients for the modeling of state-to-state energy transfer kinetics.

XI. Examples of input files and potential energy surfaces

Table 10 lists input files provided with MQCT_2023 code as examples. They cover calculations for all system types, with different modes of input (including the external data, such as pre-computed wave functions), different ways of computing (direct integration, basis set expansion) and saving the matrix (single file, formatted/unformatted, parallel IO, re-truncation), different sampling schemes (regular, steps in ℓ , Monte-Carlo sampling of ℓ, j, m), different kinds of trajectory propagation (CC, CS, AT) and analysis of the final data (total and differential cross sections, printing individual trajectories, handling resonances, etc.).

Key References:

1. “MQCT: User-ready program for calculations of inelastic scattering of two molecules” Computer Physics Communications 252, 107155 (2020).
2. “Adiabatic trajectory approximation: a new general method in the toolbox of mixed quantum/classical theory for collisional energy transfer” Journal of Physical Chemistry Letters 14, 817–824 (2023).
3. “Mixed quantum/classical theory for rotational energy exchange in symmetric-top-rotor + linear-rotor collisions and a case study of the ND₃ + D₂ system” Physical Chemistry Chemical Physics 25, 17287-17299 (2023).
4. “Description of quantum interference using mixed quantum/classical theory of inelastic scattering” Physical Chemistry Chemical Physics 25, 15683-15692 (2023).
5. “MQCT 2024: A program for calculations of inelastic scattering of two molecules (new version announcement)” Computer Physics Communications 294, 108938 (2024).
6. “Mixed quantum/classical theory (MQCT) approach to the dynamics of molecule–molecule collisions in complex systems” Faraday Discussions 251, 225-248 (2024).
7. “Mixed quantum/classical theory for rotationally inelastic scattering of identical collision partners revised” Physical Chemistry Chemical Physics 26, 27567-27582 (2024).

Table 10: A list of potential energy surfaces and example input files for different system types provided with the MQCT_2026

SYS_TYPE	PES Included	Name of the input file	Descriptions of examples provided
1	N ₂ + O	N2_O_Single_Run.inp	Compute matrix and propagate trajectories a single run, and differential cross sections
	CO + He	CO_He_mtrx_dir_int.inp CO_He_dynamics.inp	Compute matrix by direct integration in the first run, then propagate trajectories in the separate second run
2	CO + H	CO_H_Morse.inp	Use Morse parameters to construct vibrational wavefunctions of a non-rotating Morse oscillator in the ro-vibrational calculations (an approximate treatment).
		CO_H_external_vib_wfn.inp	Compute matrix using ro-vibrational energies and wavefunctions supplied externally as USER_DEFINED_BASIS.DAT
3	C ₆ H ₆ + He	C6H6_He_exp_term_list.inp	Generate the list of expansion terms in the file EXPANSION_TERMS_LIST.DAT
		C6H6_He_AT_MQCT_1st_run.inp C6H6_He_AT_MQCT_2nd_run.inp	Use AT_MQCT decoupling approximation for propagation of trajectories in two runs with constant time-step RK4 propagator
		H2O_He_CalcExp.inp	Compute expansion coefficients and save to PES_EXPAN_TERMS.DAT
4	H ₂ O + He	H2O_He_external_rot_wfn.inp	Compute matrix elements using rotational states (energies and wave function expansions) supplied externally in the input file USER_DEFINED_BASIS.DAT
	HCOOCH ₃ + He	MF_He_CS_MQCT.inp	Propagate trajectories using coupled states approximation, CS-MQCT
5	CO + CO	CO_CO_MTRX_using_exp.inp	Compute matrix elements for indistinguishable collision partners using PES expansion
		CO_CO_Resonance.inp	Identify and print resonance trajectories using NMB_LOOPS and NMB_OSCIL
6	CO + H ₂	CO_H2_Parallel_IO.inp	Compute and save matrix elements using the PARALLEL_IO keyword
		CO_H2_Prob_spline.inp	Read matrix elements using PARALLEL_IO; use DL for orbital angular momentum ℓ to skip trajectories and then use PROB_SPLINE to interpolate probabilities
7	ND ₃ + D ₂	ND3_D2_VGridFile.inp	Compute VGRID_UF.DAT to store the PES at the points of the grid
		ND3_D2_IR_BGN_FIN.inp	Calculate PES_EXPAN_TERMS.DAT in several runs for different parts of R-grid
8	H ₂ O + H ₂	H2O_H2_exp_term.inp	Generate the list of expansion terms in the file EXPANSION_TERMS_LIST.DAT
		H2O_H2_Parallel_IO.inp	Compute, save and read matrix elements using the PARALLEL_IO keyword
9	H ₂ O + NH ₃ (pairwise LJ)	H2O_NH3_MTRX_Retruncation.inp	Use MIJ_CUTOFF keyword to re-truncate and save the matrix of smaller size
0	H ₂ O + H ₂ O	H2O_H2O_AT_MQCT_1st_run.inp H2O_H2O_AT_MQCT_2nd_run.inp	Use AT_MQCT decoupling approximation for propagation of trajectories in two runs using ADAPTOL
		H2O_H2O_Identical.inp	Calculate matrix elements for identical collisional partners using keyword IDENTICAL_PES symmetrized expansion of potential energy surface
		H2O_H2O_MonteCarlo.inp	Compute cross sections using random sampling of the initial conditions

Note: Examples for SYS_TYPE=6, 8, 9, 0 include externally provided wavefunctions. Examples for SYS_TYPE=1 includes EMAX, and JMAX, and SYS_TYPE=2 includes NMB_ENRGS. Also, SYS_TYPE=3 includes AXL_SYM1 and EQU_SYM1, and SYS_TYPE=8 includes ODD_L1L2 optional keywords.